SingleRAN

CPRI MUX (LampSite) Feature Parameter Description

Issue Draft A

Date 2025-12-31

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue does not introduce any changes to SRAN21.1 02 (2025-08-30).

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to UMTS, LTE FDD, LTE TDD, NB-IoT, NR, and RFA.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB.

Unless otherwise specified, in this document, NR and gNodeB always include FDD and TDD. In scenarios where they need to be distinguished, NR FDD and NR TDD are used. The same rules apply to gNodeB.

The"U", "L", "T", "C", "M", "N", "N (TDD)", and "N (FDD)" in RAT acronyms refer to UMTS, LTE FDD, LTE TDD, CDMA, LTE NB-IoT, NR, NR TDD, and NR FDD, respectively.

2.3 Differences Between Base Station Types

This document applies to DBS3900 LampSite and DBS5900 LampSite base stations. This document does not apply to macro base stations. For details about CPRI MUX applied on macro base stations, see *CPRI MUX Feature Parameter Description*.

2.4 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
CPRI MUX (LampSite)	None	**4 CPRI MUX (LampSite)**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
CPRI MUX (LampSite)	Supported only in low frequency bands	**4 CPRI MUX (LampSite)**

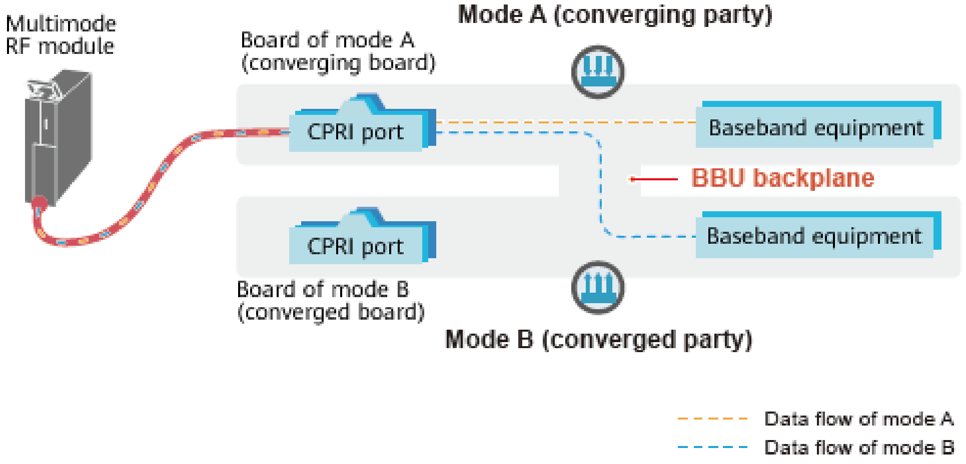
3 Overview

The CPRI MUX feature enables data transmission of multiple modes on a common public radio interface (CPRI) port.

Figure 3-1 shows CPRI MUX principles when data of multiple network modes is converged. The board of network mode B is connected to the board of network mode A through the backplane. This way, data of network mode B is converged on the board of network mode A. Then, the board of network mode A is connected to a multimode RF module through a CPRI optical fiber which carries data of both network modes A and B.

The board of network mode A serves as the converging board and network mode A is the converging party. The board of network mode B serves as the converged board and network mode B is the converged party.

Figure 3-1 CPRI MUX principles when data of multiple network modes is converged



4 CPRI MUX (LampSite)

4.1 Principles

4.1.1 Boards Supporting CPRI MUX and Constraints

This section describes boards and RF modules that support CPRI MUX. All pRRU, RHUB, and RRU models in the LampSite solution support CPRI MUX.

NOTE

This chapter describes the support and restrictions of different types of BBUs for CPRI MUX. For details about the BBU and boards in the BBU, see *LampSite BBU Hardware Description*.

4.1.1.1 BBU3900 Boards Supporting CPRI MUX and Constraints

Co-MPT

- Topology constraints

  The constraints on CPRI MUX used with a BBU3900 in co-MPT scenarios are listed as follows:

  - If the converging or converged party is UMTS, a UBBP_U*x*(1) must be installed in slot 3 or 2.
  - If a co-BBP board is configured in a BBU, and network modes involved in CPRI MUX are the same as or included in the network modes of the co-BBP board, the co-BBP board provides CPRI ports. For example, if a UBBP_UL is configured in a BBU, the UL CPRI ports are provided by the UBBP_UL, not by a UMTS only or LTE only baseband processing unit.

    NOTE

    (1) UBBP_U*x* indicates a UBBP configured with UMTS baseband resources.

- Boards and their slot constraints

  Table 4-1 describes BBU3900 boards that can serve as the converging board of CPRI MUX in co-MPT scenarios and their slot constraints.

See **Topology constraints** for BBU3900 boards that can serve as the converged board of CPRI MUX in co-MPT scenarios and their slot constraints.

Table 4-1 BBU3900 boards that can serve as the converging board of CPRI MUX in a co-MPT multimode base station and their slot constraints

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
UL	UBBP_U/L/UL	Slot 2 or 3
In this table, UBBP_U/L/UL indicates that this UBBP works in U/L/UL mode. For example, a UBBP supporting ULN but working in UL mode is expressed as UBBP_UL.		

Separate-MPT

Topology constraints

The constraints on CPRI MUX used with a BBU3900 in separate-MPT scenarios are listed as follows:

- Data convergence is allowed between two modes deployed in the same BBU. The converging party cannot serve as the converged party. Data of a network mode can only be converged to one board at a time.
- Each BBU provides a maximum of six ports for data convergence. These ports must be located on the same UBBP_U/UBBP_L.
- When UMTS serves as the converged party, one UBBP_U must be installed in slot 2 or 3. When LTE serves as the converged party, one UBBP_L must be installed in slot 2 or 3.

Boards and their slot constraints

Table 4-2 describes BBU3900 boards that support CPRI MUX in separate-MPT scenarios and their slot constraints.

Table 4-2 BBU3900 boards that support CPRI MUX in a separate-MPT multimode base station and their slot constraints

Network Mode Served by the CPRI Port	Converging Party	Converging Board	Slot of the Converging Board	Converged Board	Slot of the Converged Board
U+L	UMTS	UBBP_U	Slot 2 or 3	UBBP_L	One UBBP_L must be installed in slot 3 or 2.
	LTE	UBBP_L	Slot 2 or 3	UBBP_U	One UBBP_U must be installed in slot 3 or 2.
In this table, UBBP_U/L/UL indicates that this UBBP works in U/L/UL mode. For example, a UBBP supporting ULN but working in UL mode is expressed as UBBP_UL.					

4.1.1.2 BBU3910 Boards Supporting CPRI MUX and Constraints

Co-MPT

In co-MPT scenarios, either or both UMTS and LTE can serve as the converging party or converged party. **Table 4-3** lists CPRI MUX-capable BBU3910 boards and slot constraints.

A new function introduced in SRAN11.0 supports LTE FDD and TDD co-MPT. This allows for UL or FT co-MPT.

In SRAN12.1, SRAN13.1 and later versions, RFA is introduced as a new network mode. To support RFA, the MERC is added in co-MPT scenarios to serve as a converged board.

Table 4-3 BBU3910 boards that support CPRI MUX in a co-MPT multimode base station and slot constraints

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
UL	UBBPd/e_U/L	Slots 0 to 5
ULR	UBBPd/e_U/L	Slots 0 to 5
FT	UBBPd/e_L	Slots 0 to 5

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
FTR	UBBPd/e_L	Slots 0 to 5
ULRN[a]	UBBP_N	Slots 0 to 5
a: In a co-MPT multimode base station where NR is included, optical fiber connections must be provided by the board working in NR mode. In the table, the ULRN combination represents all mode subsets including NR.		

Starting from SRAN11.0, slots 2 and 3 in the BBU3910 can be used at the same time to provide optical fiber connections in UL mixed operation.

In SRAN12.1, SRAN13.1 and later versions, slots 2 and 3 in the BBU3910 can be used at the same time to provide optical fiber connections in ULR mixed operation.

Table 4-4 BBU3910 boards that support two-slot CPRI MUX cabling in a co-MPT multimode base station

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
UL	UBBPd/e_U/L	Slots 2 and 3
ULR	UBBPd/e_U/L	Slots 2 and 3
ULRN[a]	UBBP_N	Slots 2 and 3
a: In a co-MPT multimode base station where NR is included, optical fiber connections must be provided by the board working in NR mode. In the table, the ULRN combination represents all mode subsets including NR.		

Separate-MPT

In separate-MPT scenarios, only UMTS and LTE can serve as the converging party. UMTS, LTE and RFA can serve as the converged party. **Table 4-5** lists CPRI MUX-capable BBU3910 boards and slot constraints.

Since SRAN11.0, FT co-MPT is supported, allowing for U+FT separate-MPT.

In SRAN12.1, SRAN13.1 and later versions, only RFA and LTE can share a main control board in the BBU3910 in RUL separate-MPT networking, while UMTS alone must use a main control board. The MERC serves as the baseband processing unit for RFA. The UBBPd/e_L serves as the baseband processing unit for LTE. The UBBPd/e_U serves as the baseband processing unit for UMTS. A baseband processing unit cannot be shared across network modes. Only the LTE baseband processing unit can provide optical fiber connections.

Table 4-5 BBU3910 boards that support CPRI MUX in a separate-MPT multimode base station and slot constraints

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board	Converged Board	Slot of the Converged Board
U+L	UBBPd/e_U	Slots 0 to 5	UBBPd/e_L	Slots 0 to 5
	UBBPd/e_L	Slots 0 to 5	UBBPd/e_U	Slots 0 to 5
U+FT	UBBPd/e_U	Slots 0 to 5	UBBPd/e_L	Slots 0 to 5
	UBBPd/e_L	Slots 0 to 5	UBBPd/e_U	Slots 0 to 5
U+LR	UBBPd/e_L	Slots 0 to 5	UBBPd/e_U/MERC	Slots 0 to 5
L+N	UBBP_N	Slots 0 to 5	UBBP_L	Slots 0 to 5
U+LN	UBBP_N/LN	Slots 0 to 5	UBBP_U/L	Slots 0 to 5
LR+N	UBBP_N	Slots 0 to 5	UBBP_L/MERC	Slots 0 to 5
U+RLN	UBBP_N/LN	Slots 0 to 5	UBBP_L/U/MERC	Slots 0 to 5

Starting from SRAN11.0, fiber connections can be made on boards housed in slots 2 and 3 in the BBU3910 for UL mixed operation. In separate-MPT scenarios, one-to-two converging is used where two converging parties are allowed but must work in the same mode.

Table 4-6 BBU3910 boards that support two-slot CPRI MUX cabling in a separate-MPT multimode base station

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board	Converged Board
U+L	UBBPd/e_U+UBBPd/e_U	Slots 2 and 3	UBBP_L
	UBBPd/e_L+UBBPd/e_L	Slots 2 and 3	UBBP_U
U+LR	UBBPd/e_L	Slots 2 and 3	UBBP_U/MERC
L+N	UBBP_N+UBBP_N	Slots 2 and 3	UBBP_L

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board	Converged Board
U+LN	UBBP_N/LN +UBBP_N/LN	Slots 2 and 3	UBBP_U/L
LR+N	UBBP_N+UBBP_N	Slots 2 and 3	UBBP_L/MERC
U+RLN	UBBP_N/LN +UBBP_N/LN	Slots 2 and 3	UBBP_L/U/MERC

4.1.1.3 BBU5900/BBU5910 Boards Supporting CPRI MUX and Constraints

Co-MPT

In co-MPT scenarios, one or more of UMTS, LTE, and NR can serve as the converging party or converged party. In SRAN12.1, SRAN13.1 and later versions, RFA is introduced as a new network mode. To support RFA, the MERC is added in co-MPT scenarios to serve as a converged board. **Table 4-7** lists the BBU5900/BBU5910 boards that support CPRI MUX and slot constraints.

Table 4-7 Typical BBU5900/BBU5910 boards that support CPRI MUX in a co-MPT multimode base station and slot constraints

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board
UL	UBBP_U/L/UL	Slots 0 to 5
FT	UBBP_L	Slots 0 to 5
ULR	UBBP_U/L/UL	Slots 0 to 5
ULRN[a]	UBBP_N	Slots 0 to 5
a: In a co-MPT multimode base station where NR is included, optical fiber connections must be provided by the board working in NR mode. In the table, the ULRN combination represents all mode subsets including NR.		

Separate-MPT

- Topology constraints

  The constraints on CPRI MUX used with a BBU5900/BBU5910 in separate-MPT scenarios are listed as follows:

  – UMTS/LTE/NR can serve as the converging party. UMTS/LTE/NR/RFA can serve as the converged party.

  – In SRAN12.1, SRAN13.1 and later versions, only RFA and LTE can share a main control board in RUL separate-MPT networking, while UMTS alone must use a main control board. The MERC serves as the baseband processing unit for RFA. The UBBPd/e_L serves as the baseband processing unit for LTE. The UBBPd/e_U serves as the baseband processing unit for UMTS. A baseband processing unit cannot be shared across network modes. Only the LTE baseband processing unit can provide optical fiber connections.
  – In UL CPRI MUX or U&[L*N] hybrid-MPT scenarios, when a UBBPh series board serves as the converging board and a UBBP_U serves as the converged party, only ports 0 to 5 on the UBBPh series board can provide optical fiber connections.

- Boards and their slot constraints

  Table 4-8 lists the BBU5900/BBU5910 boards that support CPRI MUX and slot constraints.

Table 4-8 Typical BBU5900/BBU5910 boards that support CPRI MUX in a separate-MPT multimode base station and slot constraints

Network Mode Served by the CPRI Port	Converging Board	Slot of the Converging Board	Converged Board	Slot of the Converged Board
U+L	UBBP_U	Slots 0 to 5	UBBP_L	Slots 0 to 5
	UBBP_L	Slots 0 to 5	UBBP_U	Slots 0 to 5
U+FT	UBBP_U	Slots 0 to 5	UBBP_L	Slots 0 to 5
	UBBP_L	Slots 0 to 5	UBBP_U	Slots 0 to 5
U+LR	UBBP_L	Slots 0 to 5	UBBP_U/ MERC	Slots 0 to 5
L+N	UBBP_N	Slots 0 to 5	UBBP_L	Slots 0 to 5
U+LN	UBBP_N	Slots 0 to 5	UBBP_U/L	Slots 0 to 5
LR+N	UBBP_N	Slots 0 to 5	UBBP_L/ MERC	Slots 0 to 5
U+RLN	UBBP_N/LN	Slots 0 to 5	UBBP_L/U/ MERC	Slots 0 to 5

4.1.1.4 BookBBU5901 Supporting CPRI MUX and Constraints

Topology Constraints

The constraints on CPRI MUX used with a BookBBU5901 are listed as follows:

- A BookBBU5901 supports CPRI MUX only in co-MPT scenarios and supports only UMTS, LTE, UL, and LN CPRI MUX.
- Converging and converged parties must be deployed in the same BBU. Inter-BBU CPRI MUX is not supported.
- All the three CPRI ports on the BookBBU5901 support CPRI MUX.

4.1.2 CPRI MUX Networking Example

4.1.2.1 CPRI MUX Topologies with BBU3910

This section describes CPRI MUX networking scenarios for co-MPT and separate-MPT multimode LampSite base stations with the BBU3910 used.

4.1.2.1.1 Co-MPT Scenarios

Constraints

Constraints on co-MPT CPRI MUX are as follows:

- The converging and converged parties must be deployed in the same BBU.
- If a multimode baseband processing unit is configured in a BBU, and modes involved in CPRI MUX are the same as or included in the working modes of the multimode baseband processing unit, connect the BBU and RF modules over CPRI ports on the multimode baseband processing unit. For example, if a UBBP_UL is configured in a BBU, connect the BBU and RF modules over the UL CPRI ports on the UBBP_UL instead of the UL CPRI ports on a UMTS only or LTE only baseband processing unit.

Networking Mode

This section describes the networking scenarios of co-MPT CPRI MUX for the BBU3910. Scenarios that are not provided herein are considered similar to the provided ones.

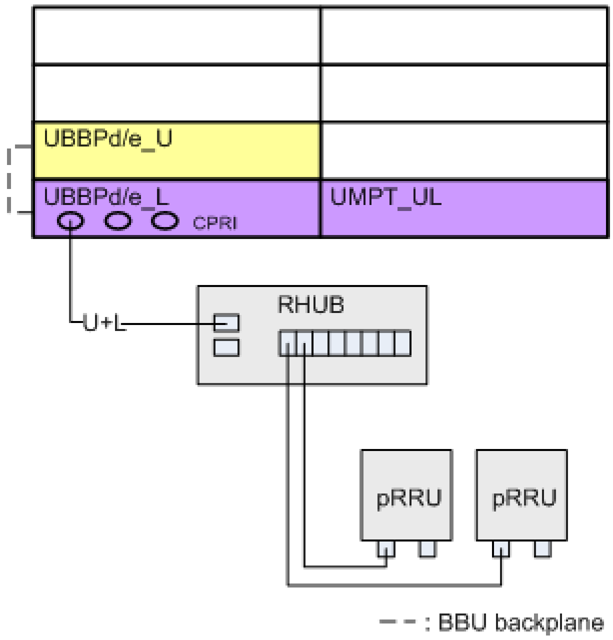
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Co-MPT CPRI MUX

Figure 4-1 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

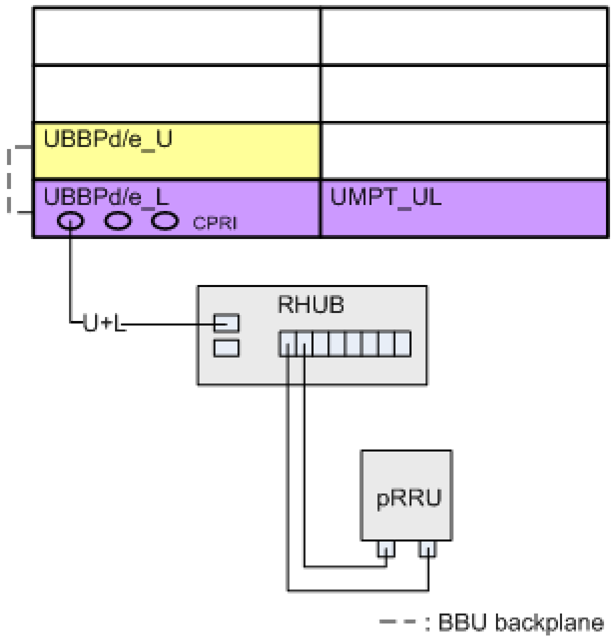
Figure 4-1 UL co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Co-MPT CPRI MUX

Figure 4-2 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

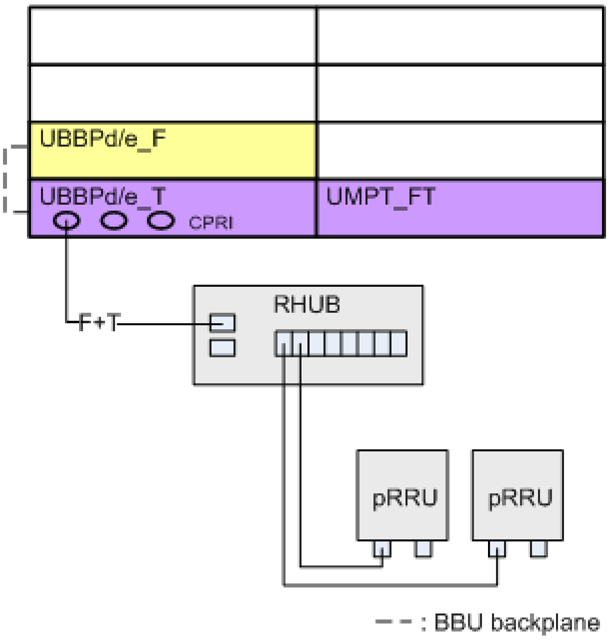
Figure 4-2 UL co-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for FT Co-MPT CPRI MUX

Figure 4-3 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

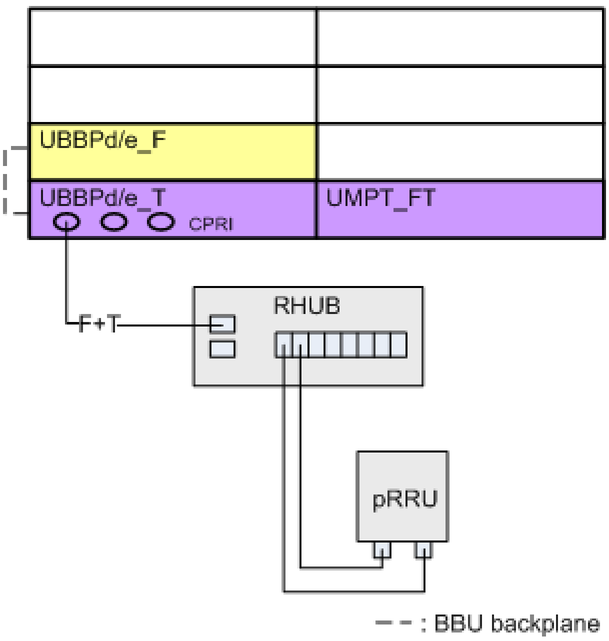
Figure 4-3 FT co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for FT Co-MPT CPRI MUX

Figure 4-4 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

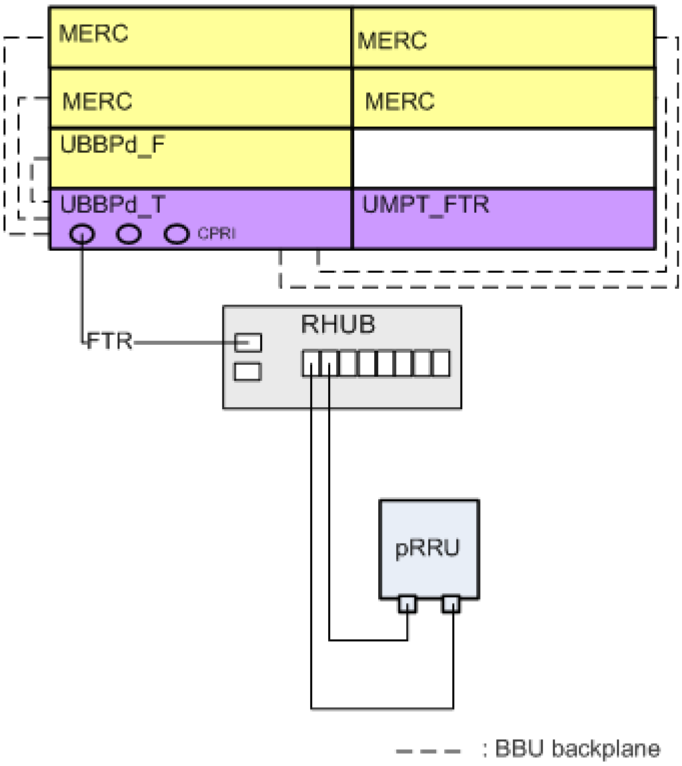
Figure 4-4 FT co-MPT CPRI MUX in branch load sharing topology



Branch Load Sharing Topology for FTR Co-MPT CPRI MUX

Figure 4-5 shows FTR co-MPT CPRI MUX, where LTE TDD serves as the converging party, and LTE FDD and RFA serve as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

Figure 4-5 FTR co-MPT CPRI MUX in branch load sharing topology



4.1.2.1.2 Separate-MPT Scenarios

Constraints

Constraints on separate-MPT CPRI MUX are as follows:

- The converging and converged parties must be deployed in the same BBU.
- Data convergence is allowed between two modes deployed in the same BBU. The converging party cannot serve as the converged party.
- When the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converging party.
- For separate-MPT CPRI MUX, a BBU provides up to 24 ports for data convergence.
- The MERC can only serve as a converged board.
- One converged board corresponds to one or two converging boards. A maximum of three pairs of one-to-one relationships or two pairs of one-to-two relationships are supported. The two-to-one convergence relationship is not supported.

Networking Mode

This section describes the networking scenarios of separate-MPT CPRI MUX for the BBU3910. Scenarios that are not provided herein are considered similar to the provided ones.

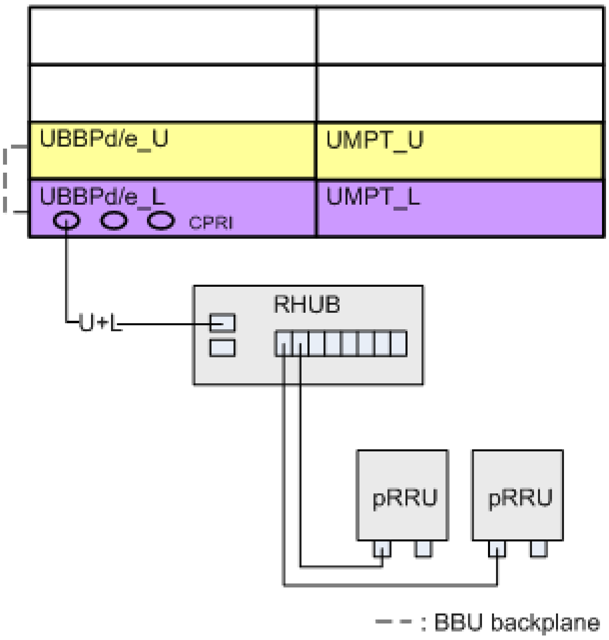
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Separate-MPT CPRI MUX

Figure 4-6 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

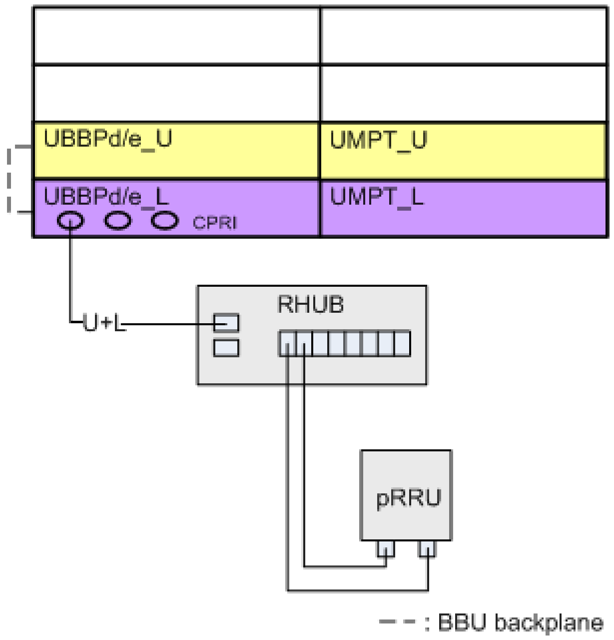
Figure 4-6 UL separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Separate-MPT CPRI MUX

Figure 4-7 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

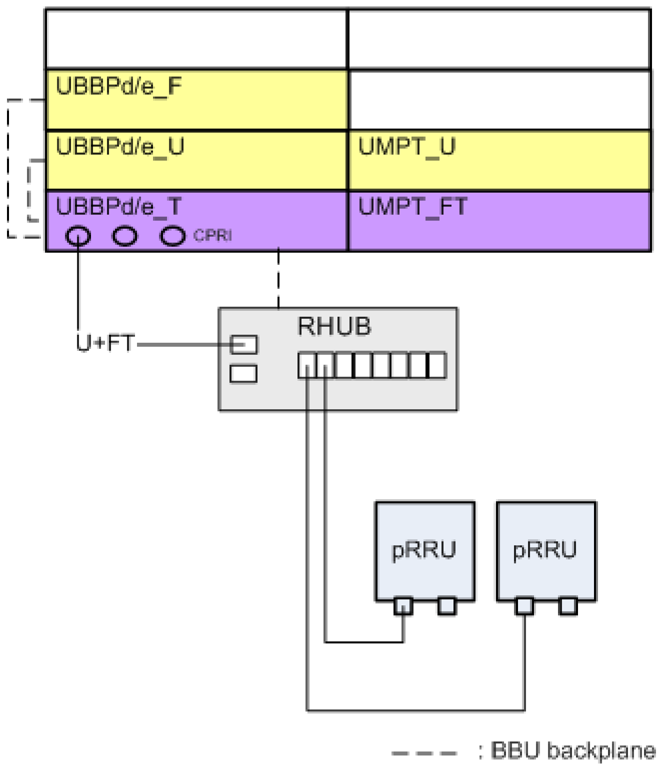
Figure 4-7 UL separate-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for U+FT Separate-MPT CPRI MUX

Figure 4-8 shows U+FT separate-MPT CPRI MUX, where LTE TDD serves as the converging party, and LTE FDD and UMTS serve as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

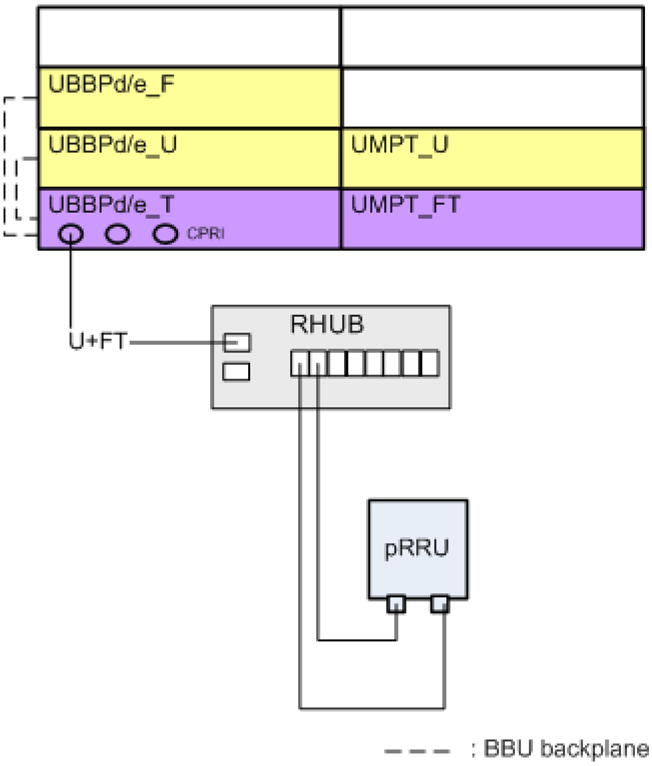
Figure 4-8 U+FT separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for U+FT Separate-MPT CPRI MUX

Figure 4-9 shows U+FT separate-MPT CPRI MUX, where LTE TDD serves as the converging party, and LTE FDD and UMTS serve as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

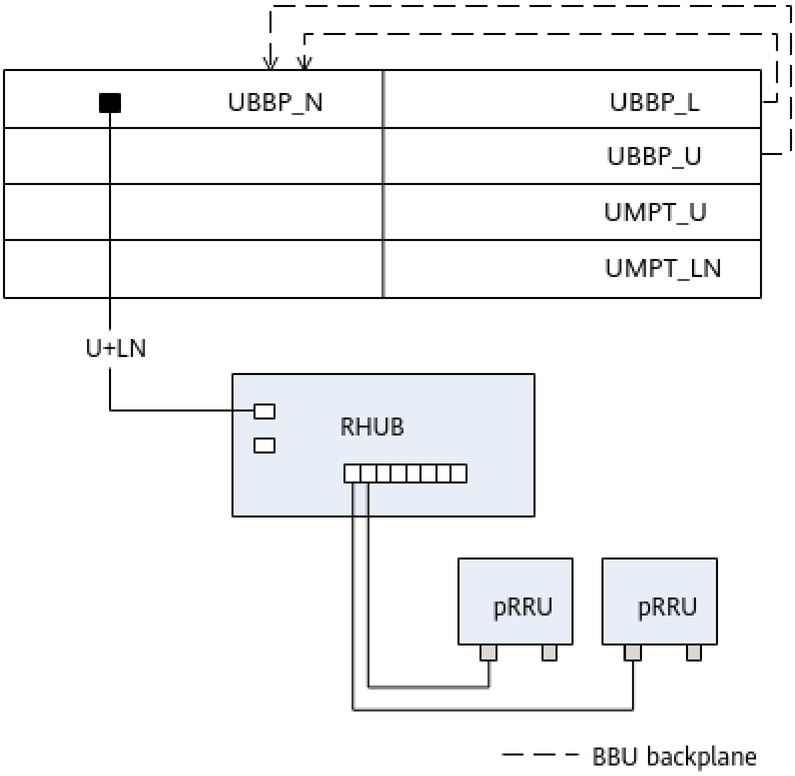
Figure 4-9 U+FT separate-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for U+LN Separate-MPT CPRI MUX

Figure 4-10 shows the topology for U&[L*N] separate-MPT CPRI MUX, where a UBBP_N serves as the converging board.

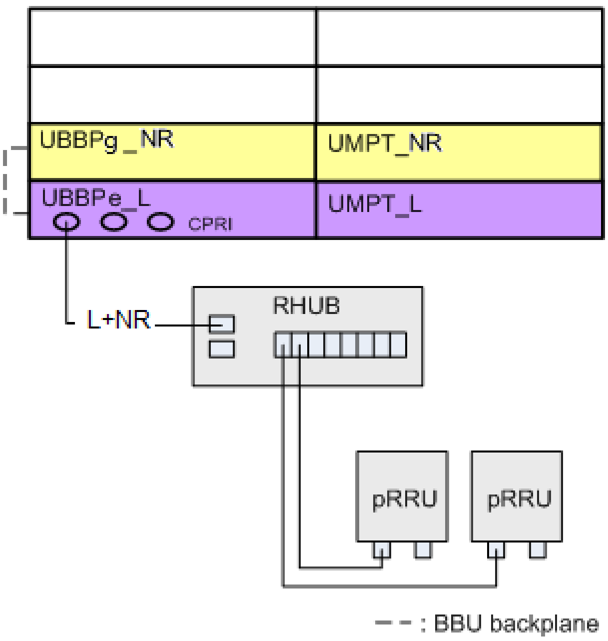
Figure 4-10 U+LN separate-MPT CPRI MUX (converging board: UBBP_N)



Branch Chain (Single-Link) Topology for LNR Separate-MPT CPRI MUX

Figure 4-11 shows LNR separate-MPT CPRI MUX, where LTE serves as the converging party and NR serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules. NR serving as the converged party also supports the load sharing topology.

Figure 4-11 LNR separate-MPT CPRI MUX in branch chain (single-link) topology



4.1.2.2 LampSite CPRI MUX Topologies with BBU5900/BBU5910

This section describes CPRI MUX networking scenarios for co-MPT and separate-MPT LampSite base stations with the BBU5900/BBU5910 used.

4.1.2.2.1 Co-MPT Scenarios

Constraints

Constraints on co-MPT CPRI MUX are as follows:

- The converging and converged parties must be deployed in the same BBU.
- The MERC can only serve as a converged board.
- If a multimode baseband processing unit is configured in a BBU, and modes involved in CPRI MUX are the same as or included in the working modes of the multimode baseband processing unit, connect the BBU and RF modules over CPRI ports on the multimode baseband processing unit. For example, if a UBBP_UL is configured in a BBU, connect the BBU and RF modules over the UL CPRI ports on the UBBP_UL instead of the UL CPRI ports on a UMTS only or LTE only baseband processing unit.

Networking Mode

This section describes the networking scenarios of co-MPT CPRI MUX for the BBU5900/BBU5910. Scenarios that are not provided herein are considered similar to the provided ones.

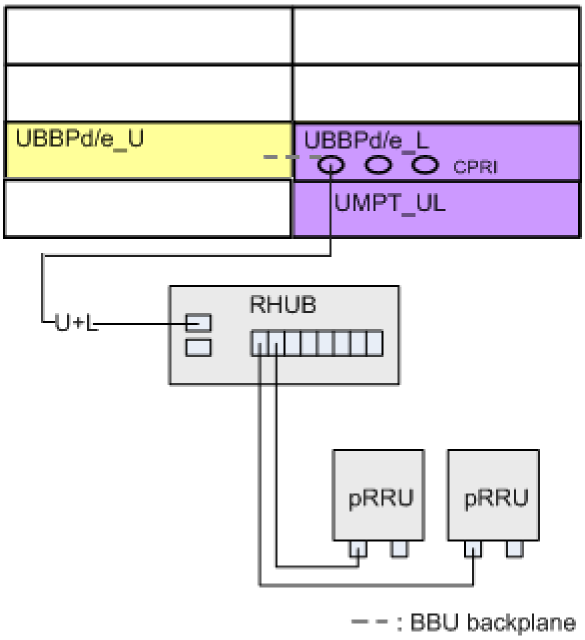
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Co-MPT CPRI MUX

Figure 4-12 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

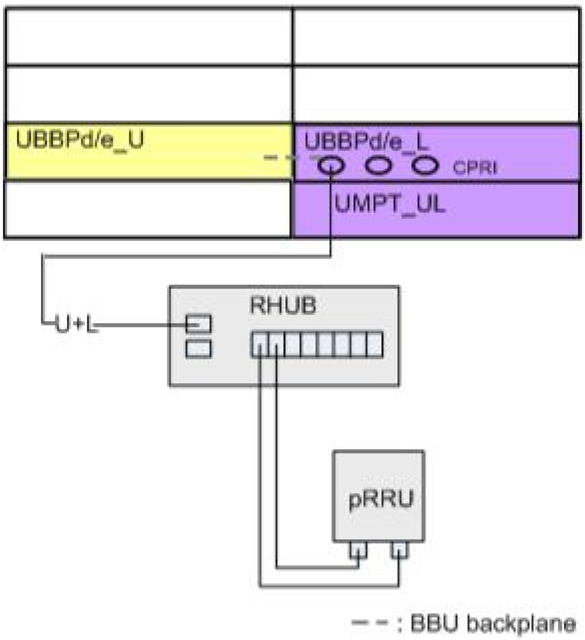
Figure 4-12 UL co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Co-MPT CPRI MUX

Figure 4-13 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

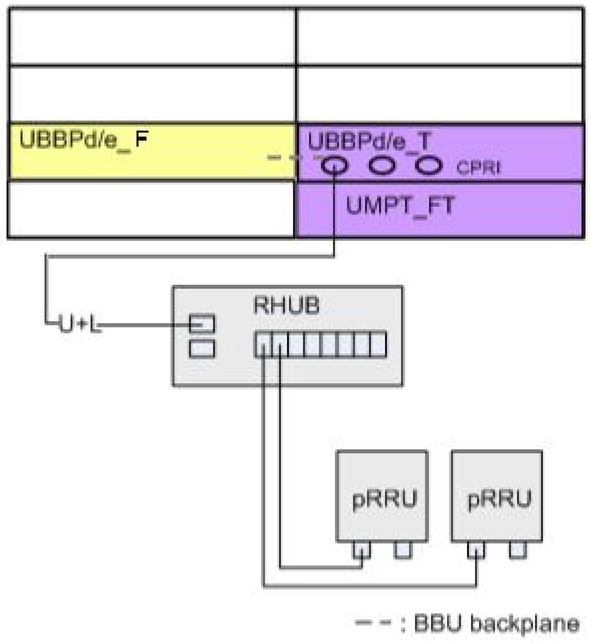
Figure 4-13 UL co-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for FT Co-MPT CPRI MUX

Figure 4-14 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

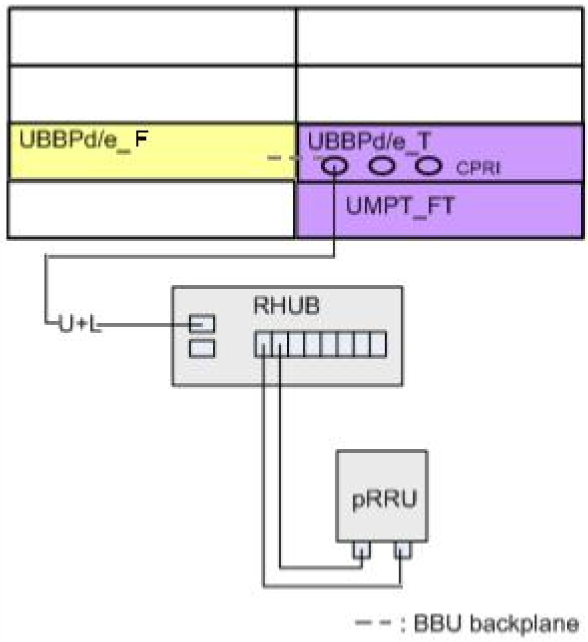
Figure 4-14 FT co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for FT Co-MPT CPRI MUX

Figure 4-15 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

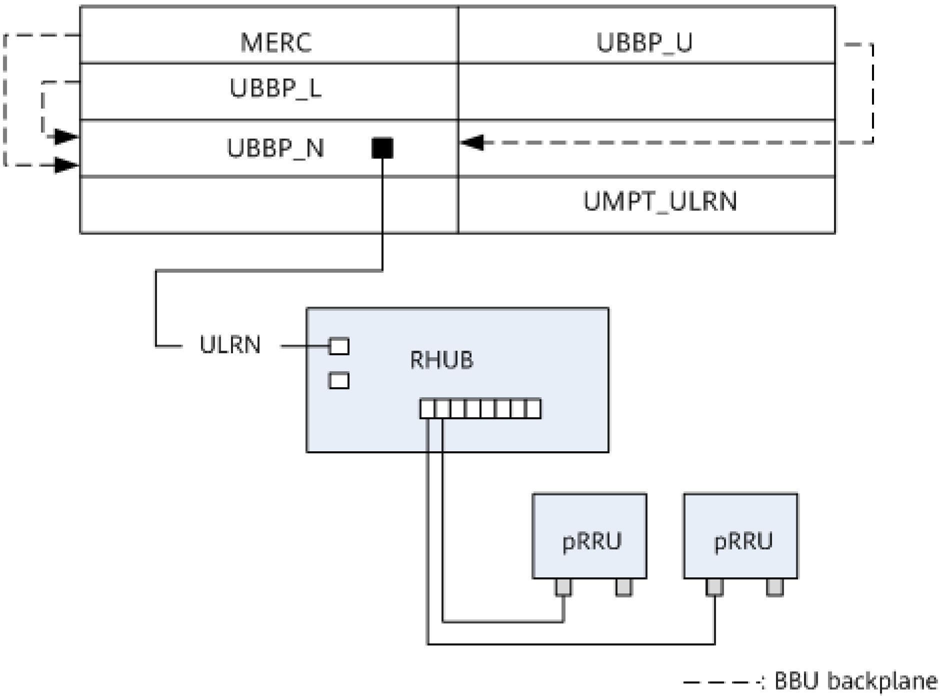
Figure 4-15 FT co-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for ULRN Co-MPT CPRI MUX

Figure 4-16 shows ULRN co-MPT CPRI MUX on a BBU5900/BBU5910, where NR serves as the converging party.

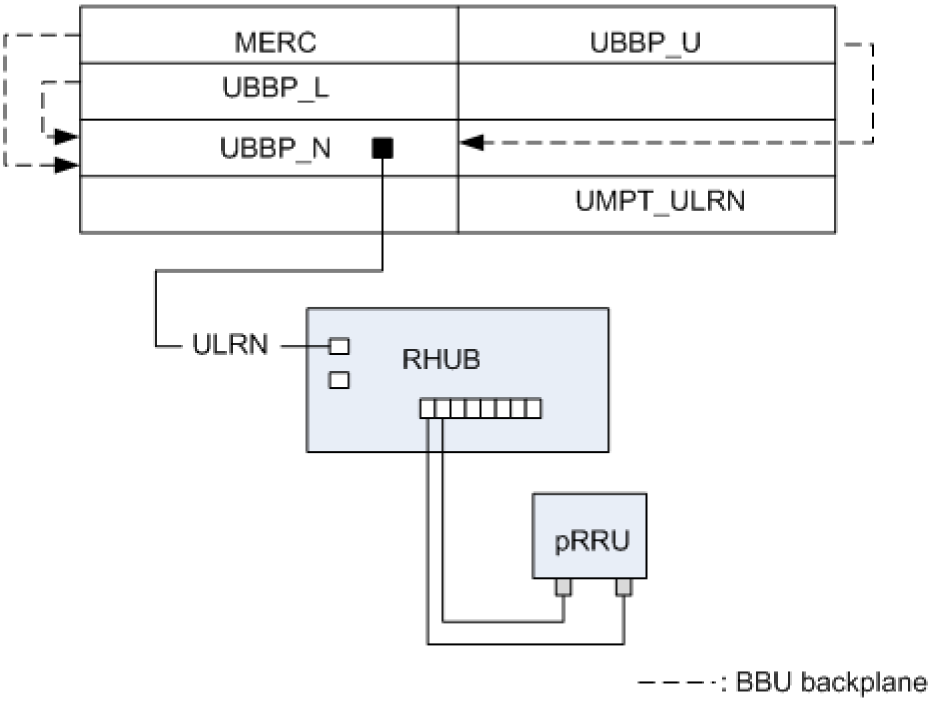
Figure 4-16 ULRN co-MPT CPRI MUX (converging board: UBBP_N)



Load Sharing Topology for ULRN Co-MPT CPRI MUX

Figure 4-17 shows ULRN co-MPT CPRI MUX on a BBU5900/BBU5910, where NR serves as the converging party.

Figure 4-17 ULRN co-MPT CPRI MUX (converging board: UBBP_N)



4.1.2.2.2 Separate-MPT Scenarios

Constraints

Constraints on separate-MPT CPRI MUX are as follows:

- The converging and converged parties must be deployed in the same BBU.
- Data convergence is allowed between two modes deployed in the same BBU. The converging party cannot serve as the converged party.
- When the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converging party.
- For separate-MPT CPRI MUX, a BBU provides up to 24 ports for data convergence.
- The MERC can only serve as a converged board.
- One converged board corresponds to one or two converging boards (one-to-one or one-to-two convergence relationships). Up to three one-to-one convergence relationships or two one-to-two convergence relationships are supported. Two-to-one convergence relationships are not supported.

Networking Mode

This section describes the networking scenarios of separate-MPT CPRI MUX for the BBU5900/BBU5910. Scenarios that are not provided herein are considered similar to the provided ones.

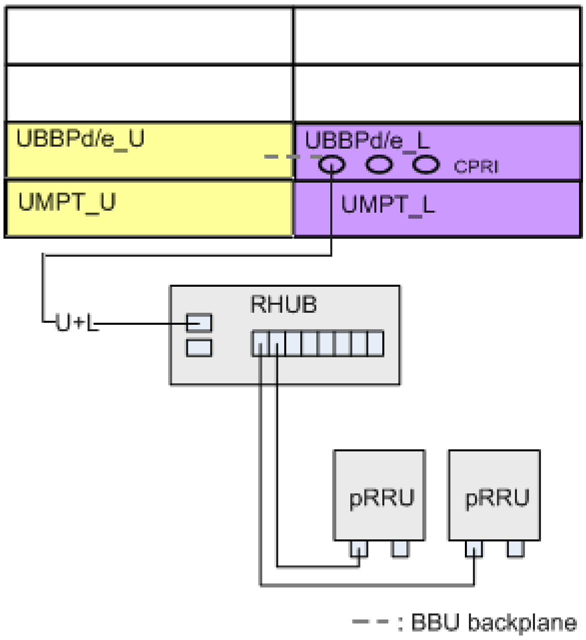
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Separate-MPT CPRI MUX

Figure 4-18 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

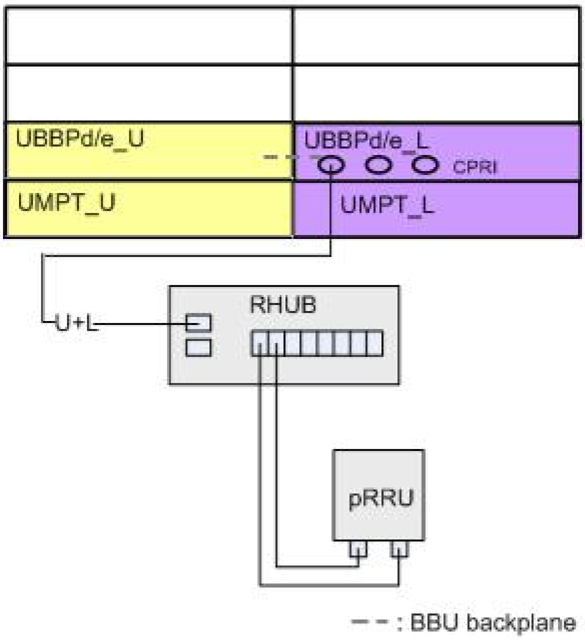
Figure 4-18 UL separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Separate-MPT CPRI MUX

Figure 4-19 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

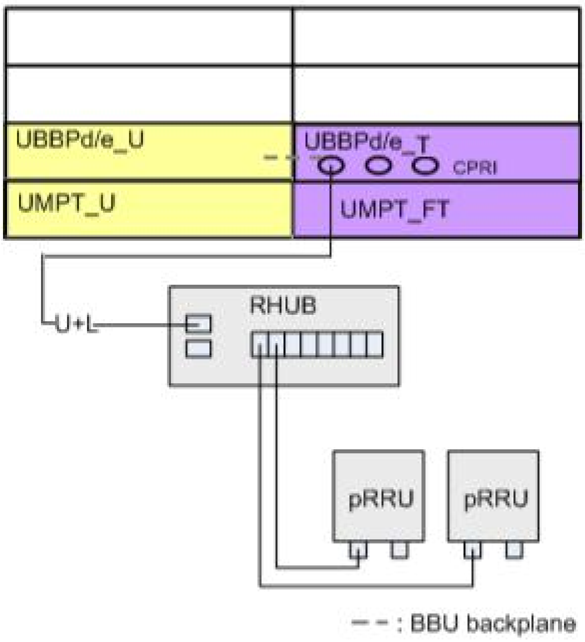
Figure 4-19 UL separate-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for U+FT Separate-MPT CPRI MUX

Figure 4-20 shows U+FT separate-MPT CPRI MUX, where LTE (FDD+TDD) serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

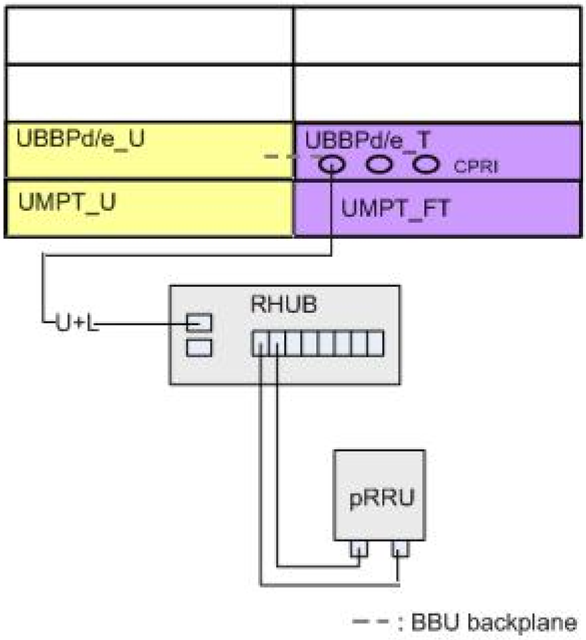
Figure 4-20 U+FT separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for U+FT Separate-MPT CPRI MUX

Figure 4-21 shows U+FT separate-MPT CPRI MUX, where LTE (FDD+TDD) serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

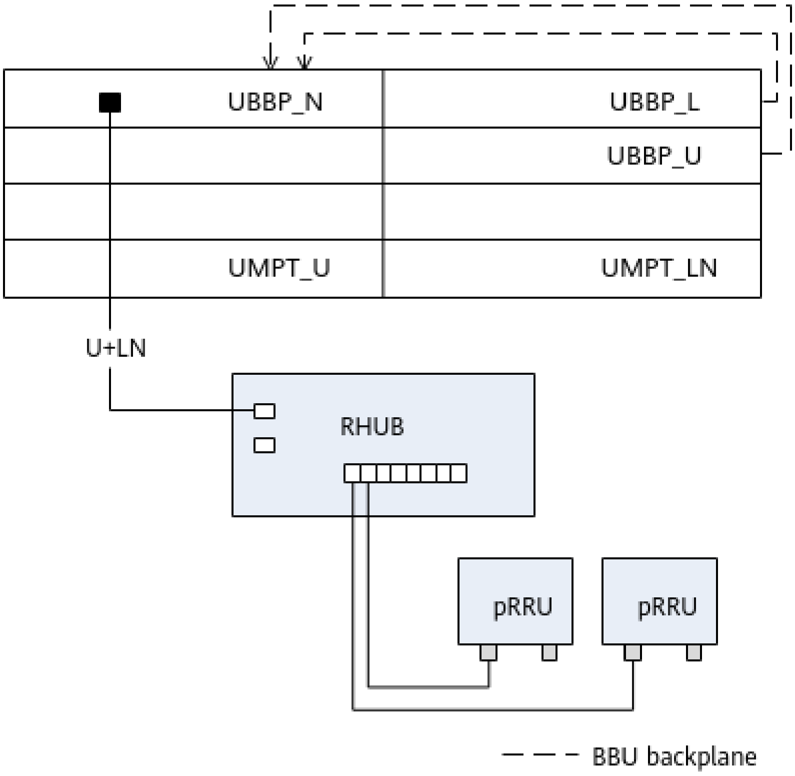
Figure 4-21 U+FT separate-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for U+LN Separate-MPT CPRI MUX

Figure 4-22 shows the topology for U&[L*N] separate-MPT CPRI MUX, where a UBBP_N serves as the converging board.

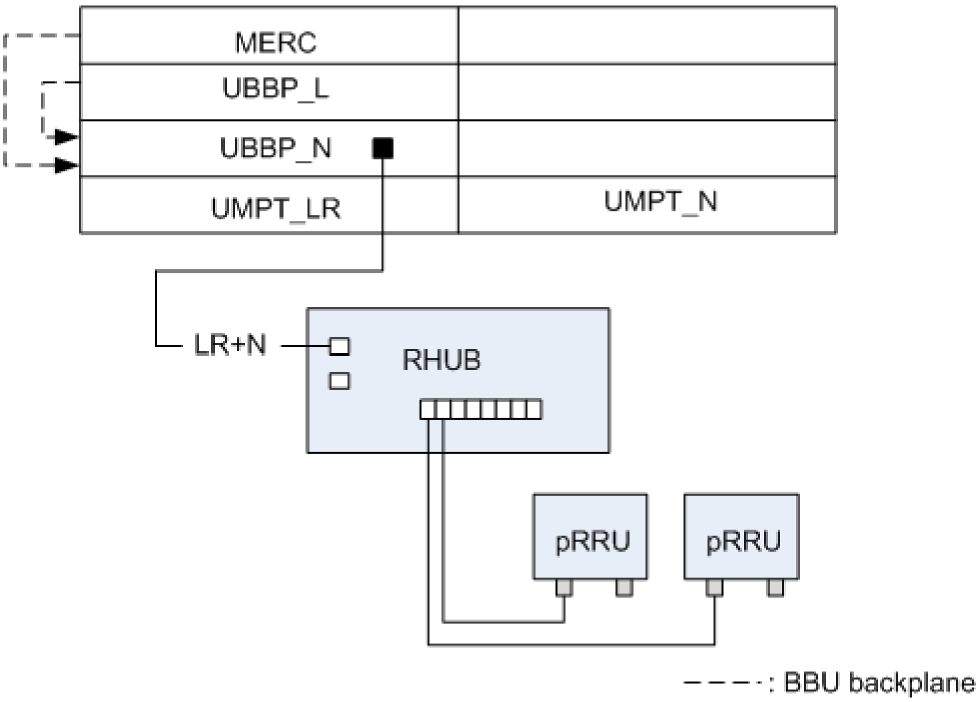
Figure 4-22 U+LN separate-MPT CPRI MUX (converging board: UBBP_N)



Branch Chain (Single-Link) Topology for LR+N Separate-MPT CPRI MUX

Figure 4-23 shows LR+N separate-MPT CPRI MUX on a BBU5900/BBU5910, where NR serves as the converging party.

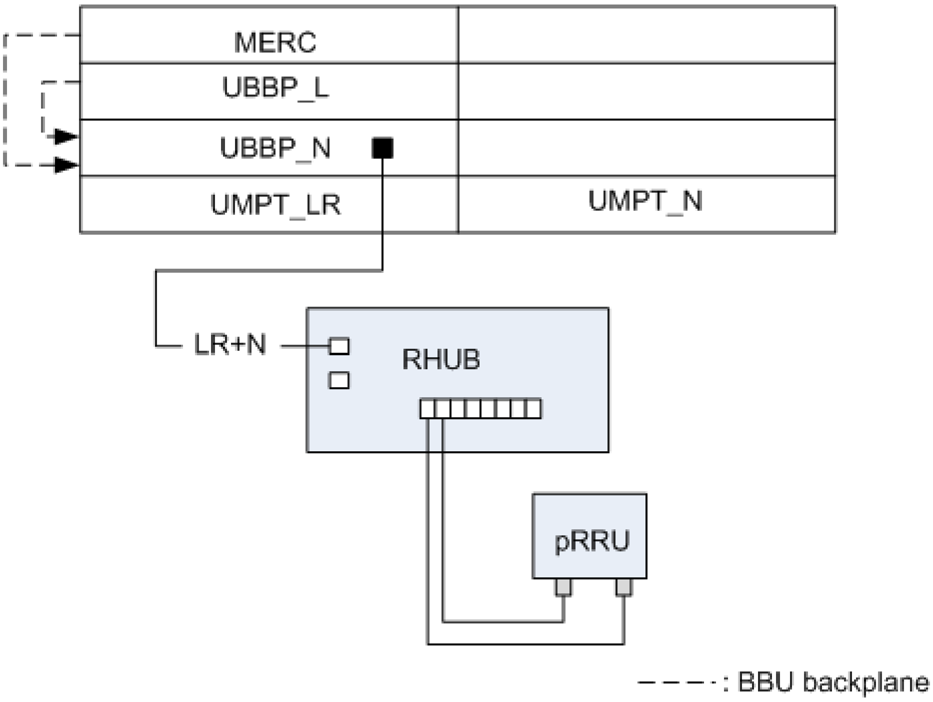
Figure 4-23 LR+N separate-MPT CPRI MUX (converging board: UBBP_N)



Branch Load Sharing Topology for LR+N Separate-MPT CPRI MUX

Figure 4-24 shows LR+N separate-MPT CPRI MUX on a BBU5900/BBU5910, where NR serves as the converging party.

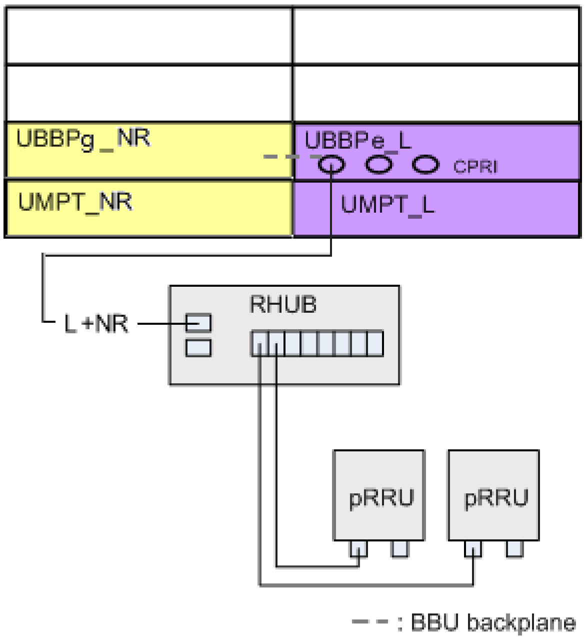
Figure 4-24 LR+N separate-MPT CPRI MUX (converging board: UBBP_N)



Branch Chain (Single-Link) Topology for LNR Separate-MPT CPRI MUX

Figure 4-25 shows LNR separate-MPT CPRI MUX, where LTE serves as the converging party and NR serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

Figure 4-25 LNR separate-MPT CPRI MUX in branch chain (single-link) topology



4.1.2.3 CPRI MUX Topologies with BBU3900

This section describes CPRI MUX networking scenarios for co-MPT and separate-MPT multimode LampSite base stations with the BBU3900 used.

4.1.2.3.1 Co-MPT Scenarios

Constraints

This networking scenario must meet the following conditions:

- The converging and converged parties must be deployed in the same BBU.
- If the converging or converged party is UMTS, a UBBP_Ux must be installed in slot 3 or 2. UBBP_Ux indicates that a UBBP configured with UMTS baseband resources.
- If a multimode baseband processing unit is configured in a BBU, and modes involved in CPRI MUX are the same as or included in the working modes of the multimode baseband processing unit, connect the BBU and RF modules over CPRI ports on the multimode baseband processing unit. For example, if a UBBP_UL is configured in a BBU, connect the BBU and RF modules over the UL CPRI ports on the UBBP_UL instead of the UL CPRI ports on a UMTS only or LTE only baseband processing unit.

Networking Mode

This section describes the networking scenarios of co-MPT CPRI MUX for the BBU3900. Scenarios that are not provided herein are considered similar to the provided ones.

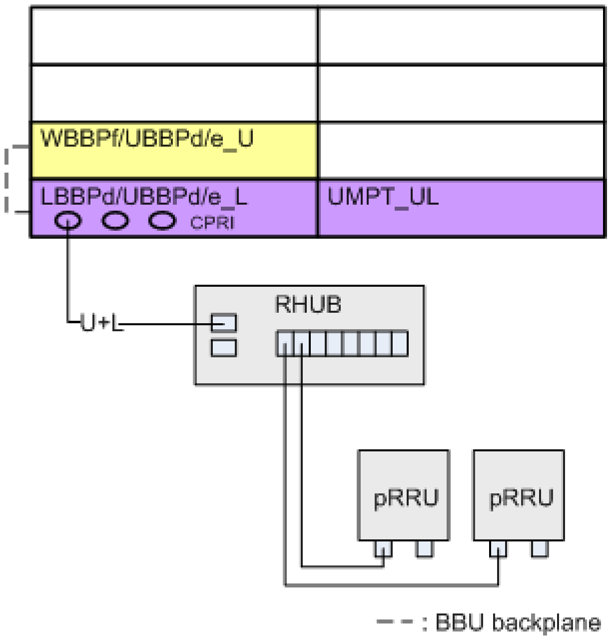
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Co-MPT CPRI MUX

Figure 4-26 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

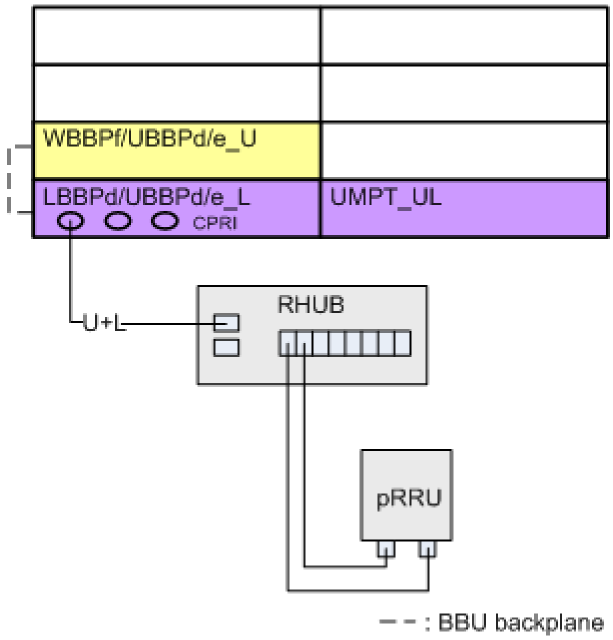
Figure 4-26 UL co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Co-MPT CPRI MUX

Figure 4-27 shows UL co-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

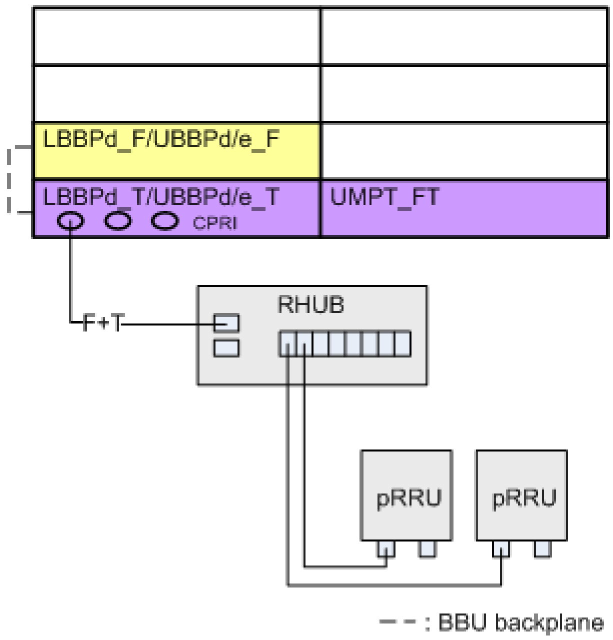
Figure 4-27 UL co-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for FT Co-MPT CPRI MUX

Figure 4-28 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

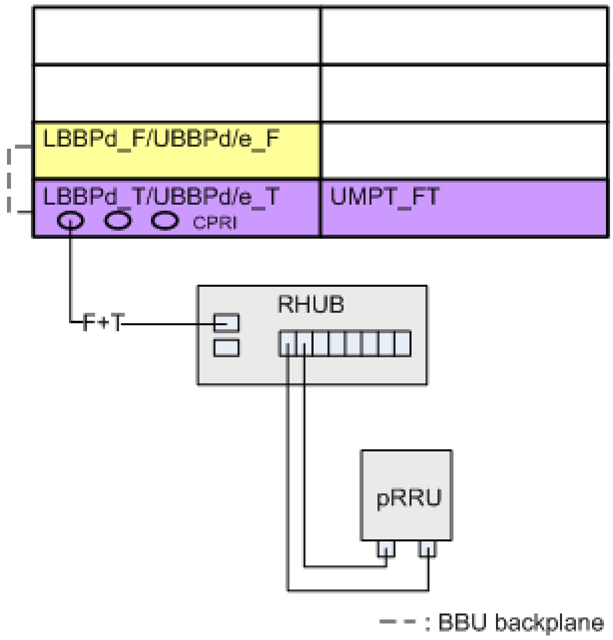
Figure 4-28 FT co-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for FT Co-MPT CPRI MUX

Figure 4-29 shows FT co-MPT CPRI MUX, where LTE TDD serves as the converging party and LTE FDD serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

Figure 4-29 FT co-MPT CPRI MUX in branch load sharing topology



4.1.2.3.2 Separate-MPT Scenarios

Constraints

Constraints on separate-MPT CPRI MUX are as follows:

- The converging and converged parties must be deployed in the same BBU.
- Data convergence is allowed between two modes deployed in the same BBU. The converging party cannot serve as the converged party.
- When the converged party supports one-to-two convergence relationships, CPRI chains/rings with **Access Type** set to **LOCALPORT** cannot be configured on the converging party.
- For separate-MPT CPRI MUX, a BBU provides up to 24 ports for data convergence.

Networking Mode

This section describes the networking scenarios of separate-MPT CPRI MUX for the BBU3900. Scenarios that are not provided herein are considered similar to the provided ones.

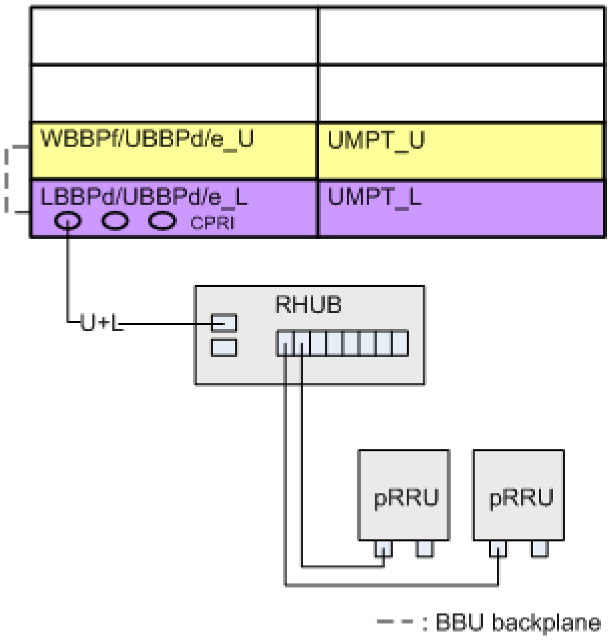
NOTE

- The networking mode of RRUs is similar to that of pRRUs. pRRU networking is exemplified herein.
- In the multimode load sharing topology, if a CPRI port or the board hosting the CPRI port is faulty, services of certain network modes may fail to be established. After the fault is rectified, services of all network modes will automatically recover.

Branch Chain (Single-Link) Topology for UL Separate-MPT CPRI MUX

Figure 4-30 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

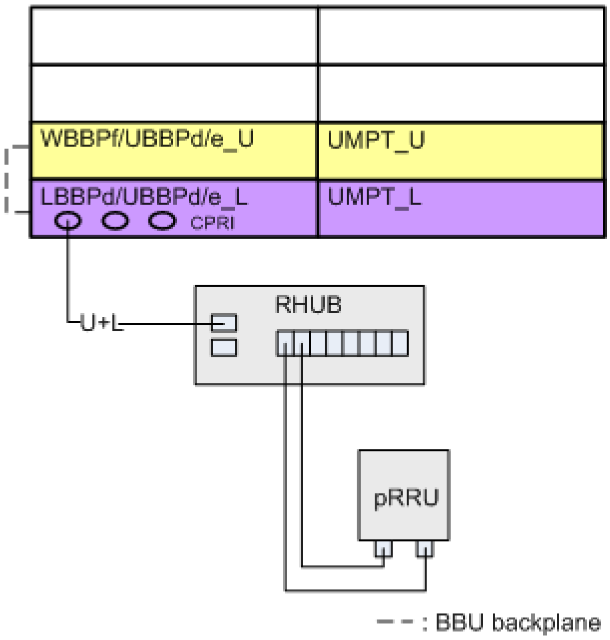
Figure 4-30 UL separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for UL Separate-MPT CPRI MUX

Figure 4-31 shows UL separate-MPT CPRI MUX, where LTE serves as the converging party and UMTS serves as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

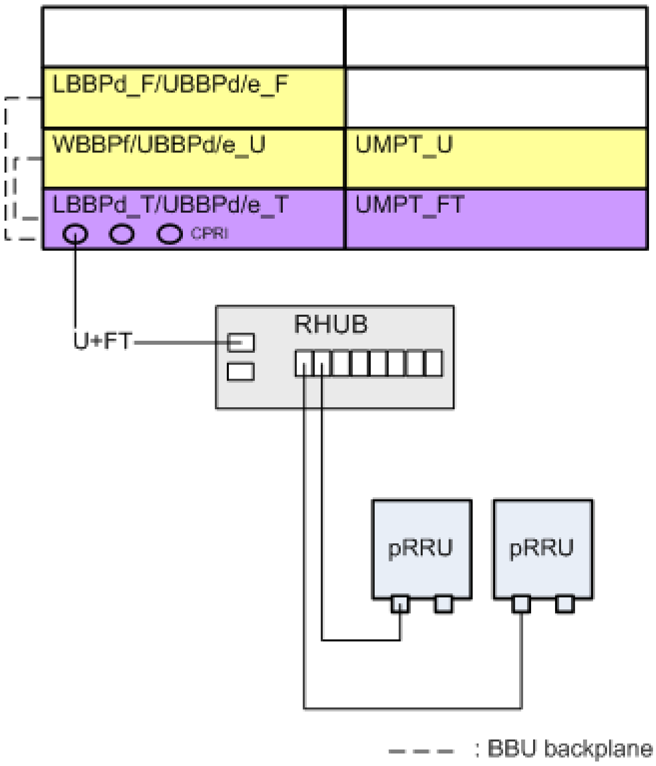
Figure 4-31 UL separate-MPT CPRI MUX in branch load sharing topology



Branch Chain (Single-Link) Topology for U+FT Separate-MPT CPRI MUX

Figure 4-32 shows U+FT separate-MPT CPRI MUX, where LTE TDD serves as the converging party, and LTE FDD and UMTS serve as the converged party. The branch chain (single-link) topology is used between the RHUB and RF modules.

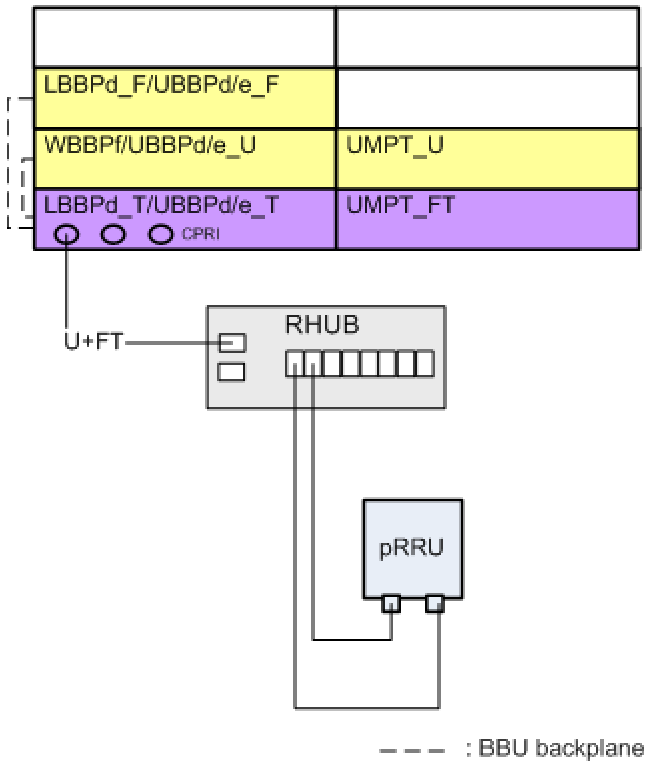
Figure 4-32 U+FT separate-MPT CPRI MUX in branch chain (single-link) topology



Branch Load Sharing Topology for U+FT Separate-MPT CPRI MUX

Figure 4-33 shows U+FT separate-MPT CPRI MUX, where LTE TDD serves as the converging party, and LTE FDD and UMTS serve as the converged party. The branch load sharing topology is used between the RHUB and RF modules.

Figure 4-33 U+FT separate-MPT CPRI MUX in branch load sharing topology



4.1.3 Specifications of CPRI MUX Capabilities

In separate-MPT or co-MPT CPRI MUX, the converging capabilities of a multimode base station are determined by CPRI line rates and board capabilities of the converging party. See **Table 4-9**, **Table 4-10**, **Table 4-11**, and **Table 4-12**.

NOTE

- In the LampSite solution, a CPRI line rate of 9.8 Gbit/s is recommended.
- The capabilities of the converging board cannot exceed the maximum carrier/cell capacity specifications of a site. For details about the maximum carrier/cell capacity specifications of a site, see "Capacity Specifications" in *LampSite BBU Technical Specifications*.
- Converging capabilities in the tables apply to common cells. For the impact of this feature working in different RATs on cell capacity, see the corresponding feature parameter description.
- UMTS-related data in the tables refers to capacity specifications when the antenna TX/RX mode is 1T2R.
- In BBU interconnection networking in U+N+L separate-MPT mode, LTE IQ data is converged on the NR baseband processing unit through the HEI port. The LTE carrier specifications must meet the bandwidth requirements of the HEI port.
- LTE includes TDD only, FDD only, and FDD+TDD. LTE TDD 5 MHz does not support compression. In applications, LTE TDD 5 MHz supports non-compression only. LTE FDD 5 MHz supports 3:2 compression. In applications, LTE FDD 5 MHz supports non-compression only and joint use of compression and non-compression.
- In the tables, "with different LTE bandwidths" indicates that there are other smaller LTE bandwidths in addition to those specified in a combination. Unless otherwise specified, a combination supports different LTE bandwidths. LampSite supports only LTE cells of 5 MHz or higher. Therefore, a combination with LTE 5 MHz includes only LTE 5 MHz and does not include other bandwidths. For instance, in the combination of 6 x LTE 20 MHz + UMTS 0 x cell (with different LTE bandwidths), LTE cells of 10 MHz or 5 MHz exist in addition to at least one LTE cell of 20 MHz.

Converging Capabilities of the Converging Party in Co-MPT CPRI MUX

Table 4-9 Converging capabilities of the converging party in multimode co-MPT base stations enabled with CPRI MUX (LTE full compression)

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
UBBPd/UBBPe	4 x LTE 20 MHz + UMTS 0 x cell (20 MHz/15 MHz only) 3 x LTE 20 MHz + UMTS 0 x cell (with different LTE bandwidths) 2 x LTE 20 MHz + UMTS 1 x cell 1 x LTE 20 MHz + UMTS 3 x cell 0 x LTE 20 MHz + UMTS 6 x cell 6 x LTE 10 MHz + UMTS 0 x cell (10 MHz only) 5 x LTE 10 MHz + UMTS 0 x cell (with different LTE bandwidths) 3 x LTE 10 MHz + UMTS 1 x cell 2 x LTE 10 MHz + UMTS 2 x cell 1 x LTE 10 MHz + UMTS 3 x cell 0 x LTE 10 MHz + UMTS 6 x cell	8 x LTE 20 MHz + UMTS 0 x cell (20 MHz/15 MHz only) 6 x LTE 20 MHz + UMTS 0 x cell (with different LTE bandwidths) 5 x LTE 20 MHz + UMTS 1 x cell 4 x LTE 20 MHz + UMTS 2 x cell 3 x LTE 20 MHz + UMTS 4 x cell 2 x LTE 20 MHz + UMTS 6 x cell 1 x LTE 20 MHz + UMTS 9 x cell 0 x LTE 20 MHz + UMTS 13 x cell 12 x LTE 10 MHz + UMTS 0 x cell (10 MHz only) 10 x LTE 10 MHz + UMTS 0 x cell (with different LTE bandwidths) 8 x LTE 10 MHz + UMTS 1 x cell 7 x LTE 10 MHz + UMTS 2 x cell 6 x LTE 10 MHz + UMTS 3 x cell 5 x LTE 10 MHz + UMTS 4 x cell 4 x LTE 10 MHz + UMTS 5 x cell 3 x LTE 10 MHz + UMTS 6 x cell 2 x LTE 10 MHz + UMTS 8 x cell

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
		1 x LTE 10 MHz + UMTS 10 x cell 0 x LTE 10 MHz + UMTS 13 x cell
Note 1: The mentioned LTE 20 MHz specifications assume a 2T2R mode, a 20 MHz carrier bandwidth, and a 2:1 compression ratio for CPRI bandwidth. The mentioned LTE 10 MHz specifications assume a 2T2R mode, a 10 MHz carrier bandwidth, and a 3:2 compression ratio for CPRI bandwidth.		

Table 4-10 Converging capabilities of the converging party in multimode co-MPT base stations enabled with CPRI MUX (LTE full non-compression or joint use of compression and non-compression)

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
UBBPd/ UBBPe	2 x LTE 20 MHz + UMTS 0 x cell 1 x LTE 20 MHz + UMTS 1 x cell 0 x LTE 20 MHz + UMTS 6 x cell 4 x LTE 10 MHz + UMTS 0 x cell (non-compression and 10 MHz only) 3 x LTE 10 MHz + UMTS 0 x cell (compression and non-compression with different LTE bandwidths) 2 x LTE 10 MHz + UMTS 1 x cell 1 x LTE 10 MHz + UMTS 3 x cell 0 x LTE 10 MHz + UMTS 6 x cell 8 x LTE 5 MHz + UMTS 0 x cell (non-compression only) 6 x LTE 5 MHz + UMTS 0 x cell (compression and non-compression) 4 x LTE 5 MHz + UMTS 1 x cell 3 x LTE 5 MHz + UMTS 2 x cell 2 x LTE 5 MHz + UMTS 3 x cell 0 x LTE 5 MHz + UMTS 6 x cell	4 x LTE 20 MHz + UMTS 0 x cell 3 x LTE 20 MHz + UMTS 1 x cell (with different LTE bandwidths) 2 x LTE 20 MHz + UMTS 3 x cell 1 x LTE 20 MHz + UMTS 6 x cell 0 x LTE 20 MHz + UMTS 13 x cell 8 x LTE 10 MHz + UMTS 0 x cell (10 MHz only) 6 x LTE 10 MHz + UMTS 0 x cell (with different LTE bandwidths) 5 x LTE 10 MHz + UMTS 2 x cell 4 x LTE 10 MHz + UMTS 3 x cell 3 x LTE 10 MHz + UMTS 5 x cell 2 x LTE 10 MHz + UMTS 6 x cell 1 x LTE 10 MHz + UMTS 9 x cell 0 x LTE 10 MHz + UMTS 13 x cell 16 x LTE 5 MHz + UMTS 0 x cell 13 x LTE 5 MHz + UMTS 1 x cell 10 x LTE 5 MHz + UMTS 2 x cell 8 x LTE 5 MHz + UMTS 3 x cell 7 x LTE 5 MHz + UMTS 4 x cell 5 x LTE 5 MHz + UMTS 6 x cell 4 x LTE 5 MHz + UMTS 7 x cell

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
		2 x LTE 5 MHz + UMTS 8 x cell 1 x LTE 5 MHz + UMTS 10 x cell 0 x LTE 5 MHz + UMTS 13 x cell
Note 1: The mentioned LTE 20 MHz specifications assume a 2T2R mode, a 20 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 2:1 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 2:1 compression are jointly used. The mentioned LTE 10 MHz specifications assume a 2T2R mode, a 10 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 3:2 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 3:2 compression are jointly used. The mentioned LTE 5 MHz specifications assume a 2T2R mode, a 5 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 3:2 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 3:2 compression are jointly used.		

Converging Capabilities of the Converging Party in Separate-MPT CPRI MUX

Table 4-11 Converging capabilities of the converging party in multimode separate-MPT base stations enabled with CPRI MUX (LTE full compression)

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
UBBPd/UBBPe	4 x LTE 20 MHz + UMTS 0 x cell (20 MHz/15 MHz only) 3 x LTE 20 MHz + UMTS 0 x cell (with different LTE bandwidths) 2 x LTE 20 MHz + UMTS 2 x cell 1 x LTE 20 MHz + UMTS 5 x cell 0 x LTE 20 MHz + UMTS 6 x cell 6 x LTE 10 MHz + UMTS 0 x cell (10 MHz only) 5 x LTE 10 MHz + UMTS 0 x cell (with different LTE bandwidths) 4 x LTE 10 MHz + UMTS 1 x cell 3 x LTE 10 MHz + UMTS 2 x cell 2 x LTE 10 MHz + UMTS 4 x cell 1 x LTE 10 MHz + UMTS 5 x cell 0 x LTE 10 MHz + UMTS 6 x cell	8 x LTE 20 MHz + UMTS 0 x cell (20 MHz/15 MHz only) 6 x LTE 20 MHz + UMTS 0 x cell (with different LTE bandwidths) 5 x LTE 20 MHz + UMTS 3 x cell 4 x LTE 20 MHz + UMTS 5 x cell 3 x LTE 20 MHz + UMTS 7 x cell 2 x LTE 20 MHz + UMTS 9 x cell 1 x LTE 20 MHz + UMTS 11 x cell 0 x LTE 20 MHz + UMTS 13 x cell 12 x LTE 10 MHz + UMTS 0 x cell (10 MHz only) 10 x LTE 10 MHz + UMTS 0 x cell (with different LTE bandwidths) 9 x LTE 10 MHz + UMTS 1 x cell 8 x LTE 10 MHz + UMTS 3 x cell 7 x LTE 10 MHz + UMTS 4 x cell 6 x LTE 10 MHz + UMTS 5 x cell 5 x LTE 10 MHz + UMTS 6 x cell 4 x LTE 10 MHz + UMTS 8 x cell 3 x LTE 10 MHz + UMTS 9 x cell

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
		2 x LTE 10 MHz + UMTS 10 x cell 1 x LTE 10 MHz + UMTS 12 x cell 0 x LTE 10 MHz + UMTS 13 x cell
Note 1: The mentioned LTE 20 MHz specifications assume a 2T2R mode, a 20 MHz carrier bandwidth, and a 2:1 compression ratio for CPRI bandwidth. The mentioned LTE 10 MHz specifications assume a 2T2R mode, a 10 MHz carrier bandwidth, and a 3:2 compression ratio for CPRI bandwidth.		

Table 4-12 Converging capabilities of the converging party in multimode separate-MPT base stations enabled with CPRI MUX (LTE full non-compression or joint use of compression and non-compression)

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
UBBPd/ UBBPe	2 x LTE 20 MHz + UMTS 0 x cell 1 x LTE 20 MHz + UMTS 3 x cell 0 x LTE 20 MHz + UMTS 6 x cell 4 x LTE 10 MHz + UMTS 0 x cell (non-compression and 10 MHz only) 3 x LTE 10 MHz + UMTS 0 x cell (compression and non-compression with different LTE bandwidths) 2 x LTE 10 MHz + UMTS 2 x cell 1 x LTE 10 MHz + UMTS 5 x cell 0 x LTE 10 MHz + UMTS 6 x cell 8 x LTE 5 MHz + UMTS 0 x cell (non-compression only) 6 x LTE 5 MHz + UMTS 0 x cell (compression and non-compression) 5 x LTE 5 MHz + UMTS 1 x cell 4 x LTE 5 MHz + UMTS 2 x cell 3 x LTE 5 MHz + UMTS 3 x cell 2 x LTE 5 MHz + UMTS 4 x cell 0 x LTE 5 MHz + UMTS 6 x cell	4 x LTE 20 MHz + UMTS 0 x cell 3 x LTE 20 MHz + UMTS 3 x cell 2 x LTE 20 MHz + UMTS 6 x cell 1 x LTE 20 MHz + UMTS 10 x cell 0 x LTE 20 MHz + UMTS 13 x cell 8 x LTE 10 MHz + UMTS 0 x cell (non-compression and 10 MHz only) 6 x LTE 10 MHz + UMTS 0 x cell (compression and non-compression with different LTE bandwidths) 5 x LTE 10 MHz + UMTS 3 x cell 4 x LTE 10 MHz + UMTS 5 x cell 3 x LTE 10 MHz + UMTS 7 x cell 2 x LTE 10 MHz + UMTS 9 x cell 1 x LTE 10 MHz + UMTS 11 x cell 0 x LTE 10 MHz + UMTS 13 x cell 16 x LTE 5 MHz + UMTS 0 x cell (non-compression only) 12 x LTE 5 MHz + UMTS 0 x cell (compression and non-compression) 11 x LTE 5 MHz + UMTS 1 x cell 10 x LTE 5 MHz + UMTS 2 x cell 9 x LTE 5 MHz + UMTS 3 x cell

Board Providing Optical Fiber Connections	Converging Capabilities of a CPRI Link (4.9 Gbit/s)	Converging Capabilities of a CPRI Link (9.8 Gbit/s)
		8 x LTE 5 MHz + UMTS 4 x cell 7 x LTE 5 MHz + UMTS 5 x cell 6 x LTE 5 MHz + UMTS 6 x cell 5 x LTE 5 MHz + UMTS 7 x cell 4 x LTE 5 MHz + UMTS 8 x cell 2 x LTE 5 MHz + UMTS 11 x cell 1 x LTE 5 MHz + UMTS 12 x cell 0 x LTE 5 MHz + UMTS 13 x cell
Note 1: The mentioned LTE 20 MHz specifications assume a 2T2R mode, a 20 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 2:1 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 2:1 compression are jointly used. The mentioned LTE 10 MHz specifications assume a 2T2R mode, a 10 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 3:2 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 3:2 compression are jointly used. The mentioned LTE 5 MHz specifications assume a 2T2R mode, a 5 MHz carrier bandwidth, and non-compression only or joint use of non-compression and 3:2 compression for CPRI bandwidth. Unless otherwise stated, non-compression and 3:2 compression are jointly used.		

4.1.3.1 Calculation of Capacity Specifications

The following are methods for calculating capacity specifications for each RAT with different configurations:

- UMTS
  - The number of supported cells in 1T2R mode is the same as that of supported cells in 2T2R mode.
  - The antenna TX/RX mode is in inversely proportion to the number of supported cells. For example, the number of supported cells in 1T1R mode is twice that of supported cells in 2T2R mode.
- LTE
  - Given the same antenna TX/RX mode and CPRI line rate, the number of supported 5 MHz or 10 MHz cells is in inverse proportion to the cell bandwidth. For example, the number of supported 5 MHz cells is twice that of supported 10 MHz cells. LTE 5 MHz cells and 10 MHz cells use the same compression ratio of 3:2.
  - Given the same antenna TX/RX mode and CPRI line rate, the number of supported 15 MHz cells is the same as that of supported 20 MHz cells.

– Given the same cell bandwidth, the number of supported cells in 1T2R mode is the same as that of supported cells in 2T2R mode.

4.2 Network Analysis

4.2.1 Benefits

This feature provides the following benefits in specific scenarios:

- New deployments

  This feature considerably reduces costs on optical modules and optical fibers for newly deployed multimode base stations. Data of different modes can be transmitted over the same CPRI port.

- Refarming

  This feature reduces engineering costs and shortens service interruption. A multimode RF unit uses a single CPRI cable at the initial phase. CPRI MUX enables newly deployed modes to reuse the existing multimode RF unit and CPRI cable.

- Cascading of multimode RF unit

  This feature guarantees indoor coverage on different floors or in different areas covered by multimode RF units that are cascaded. The LampSite solution supports RHUB cascading and RRU cascading but does not support pRRU cascading.

4.2.2 Impacts

For details about the impact between the function and other functions, see **4.3.2.3 Function Impacts**.

CPRI MUX in Separate-MPT

- Operations or exceptions on the converging party: Circuit switched (CS) and packet switched (PS) services of the converged party will be interrupted when a CPRI port of the converging party becomes faulty or when any of the following operations is performed on the converging party: software reset or power-cycle reset, blocking, or removing and then inserting of the main control board or converging board, or commissioning of CPRI ports on the converging board. Most interruptions last less than 3 minutes and the longest interruption does not exceed 5 minutes.
- Operations or exceptions on the converged party
  – If the software of the main control board of the converged party is reset, the data rate of PS services of the converging party may be decreased for less than 3s in most cases. In the worst case, the data rate decrease does not last over 10s.
  – If the main control board of the converged party is reset through a power cycle or removed and then inserted, CS and PS services of the converging party will be interrupted for less than 1 minute in most cases. In the worst case, the service interruption does not last over 3 minutes.

NOTE

In LTE/NR separate-MPT scenarios, reestablishment of links between the RHUB and pRRUs will occur and consequently LTE and NR services will be interrupted for about three minutes if any of the following operations is performed:

- Run the **ADD RRU** or **MOD RRU** command to add an RF module supporting UMTS or RFA mode.
- Run the **MOD BBP** command to add a UMTS baseband processing unit.
- Run the **ADD MERC** or **RMV MERC** command to add or remove a board.

CPRI MUX in Co-MPT

- CS and PS services of the converged party will be interrupted when a CPRI port of the converging party becomes faulty or when any of the following operations is performed on the converging party: software reset or power-cycle reset, blocking, or removing and then inserting of the main control board or converging board, or commissioning of CPRI ports on the converging board. Most interruptions last less than 3 minutes and the longest interruption does not exceed 5 minutes.
- If a UBBPd/UBBPe serves as the converging board, deleting or resetting the applications of the modes configured on this board does not affect the converged party.
- Adding baseband resources to one mode on a UBBPd/UBBPe interrupts services of other modes on this board, but does not affect services of the converged party that communicates with this board through the BBU backplane.

NOTE

Service interruption duration does not include the duration of manual operations, such as blocking a board or removing and then inserting a board.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Networking

Perform network planning according to **4.1.2.3 CPRI MUX Topologies with BBU3900**, **4.1.2.1 CPRI MUX Topologies with BBU3910**, and **4.1.2.2 LampSite CPRI MUX Topologies with BBU5900/BBU5910**.

4.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

RF modules have been installed. Cables between RF modules and BBUs have been connected as required for deploying CPRI MUX. For detailed installation procedures, see related installation guide and site maintenance guide.

When selecting the target boards and RF modules for CPRI MUX, you must consider the application scenarios and constraints provided in **4.1.1 Boards Supporting CPRI MUX and Constraints**, **4.1.2.3 CPRI MUX Topologies with BBU3900**, **4.1.2.1 CPRI MUX Topologies with BBU3910**, and **4.1.2.2 LampSite CPRI MUX Topologies with BBU5900/BBU5910**.

NOTICE

- If CPRI MUX is to be deployed on a single-mode base station that is to be reconstructed into a multimode base station, the hardware must meet requirements for the CPRI MUX feature. Otherwise, it is necessary to replace the hardware.
- The CPRI line rates of baseband processing units or interface boards, optical modules, and RF modules must match each other and meet the specifications in **4.1.3 Specifications of CPRI MUX Capabilities**.

4.4 Operation and Maintenance

4.4.1 When to Use

CPRI MUX for multimode base stations is recommended in the following scenarios:

- Sites in which a single-mode base station is upgraded to a multimode base station and the operator requires that the existing topology of optical fibers remain unchanged to reduce engineering costs
- Sites in which a co-MPT dual-mode or multimode base station is to be deployed and the operator requires that costs of optical fibers be reduced

- Sites in which coverage must be ensured through cascading of multimode RF units, such as stadiums, subways, and high-speed railways

4.4.2 Data Preparation

Before deploying CPRI MUX, collect the names of baseband processing units and RF modules as well as information about optical modules and site capacity.

4.4.2.1 Data Preparation

- The following table describes the parameters that must be set in the **RRUCHAIN** MO to configure the RHUB chain.

Parameter Name	Parameter ID	Setting Notes	Data Source
Topo Type	***TT (LTE eNodeB, 5G gNodeB)***	Set this parameter to **CHAIN**.	Network plan (negotiation not required)
Backup Mode	***BM (LTE eNodeB, 5G gNodeB)***	Set this parameter to **COLD**.	Network plan (negotiation not required)

- The following table describes the parameters that must be set in the **RHUB** MO to configure the RHUB.

Parameter Name	Parameter ID	Setting Notes	Data Source
RHUB Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the actual number of the chain/ring.	Network plan (negotiation not required)
RHUB Position	***PS (LTE eNodeB, 5G gNodeB)***	This parameter specifies the position of an RHUB on the chain/ring. It is numbered from the start port in sequence. For example, set this parameter to **0** for the first-level RHUB and **1** for the second-level RHUB.	Network plan (negotiation not required)

Parameter Name	Parameter ID	Setting Notes	Data Source
RHUB Name	***RN (LTE eNodeB, 5G gNodeB)***	This parameter specifies the name of the configured RHUB.	Network plan (negotiation not required)

- The following table describes the parameters that must be set in the **RRUCHAIN** MO to configure the pRRU or RRU chain.

Parameter Name	Parameter ID	Setting Notes	Data Source
Topo Type	***TT (LTE eNodeB, 5G gNodeB)***	Set this parameter to **CHAIN**.	Network plan (negotiation not required)
Backup Mode	***BM***	Set this parameter to **COLD**. **NOTE** This parameter needs to be configured when an RHUB and the connected pRRUs work in branch chain (single-link) topology, and does not need to be configured when an RHUB and the connected pRRUs work in branch load sharing topology.	Network plan (negotiation not required)

- The following table describes the parameters that must be set in the **RRU** MO to configure the pRRU or RRU.

Parameter Name	Parameter ID	Setting Notes	Data Source
RRU Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the actual number of the chain/ring.	Network plan (negotiation not required)
RRU Position	***PS (LTE eNodeB, 5G gNodeB)***	Set this parameter to **0**.	Network plan (negotiation not required)

Parameter Name	Parameter ID	Setting Notes	Data Source
RRU type	***RT (LTE eNodeB, 5G gNodeB)***	Set this parameter to **MPMU**/**EPRRU**[a].	Network plan (negotiation not required)
Topo Position	***TP (LTE eNodeB, 5G gNodeB)***	Set this parameter to **BRANCH**.	Network plan (negotiation not required)
RF Unit Working Mode	***RS (LTE eNodeB, 5G gNodeB)***	Select the RAT combination as required.	Network plan (negotiation not required)
RRU Name	***RN (LTE eNodeB, 5G gNodeB)***	This parameter specifies the name of the configured pRRU/RRU.	Network plan (negotiation not required)
Maintenance Mode	***MNTMODE (LTE eNodeB, 5G gNodeB)***	Set this parameter to **NORMAL(NORMAL)**.	Network plan (negotiation not required)
DL Signal Quality Detection Switch	***RFTXSIGNDETECTSW (LTE eNodeB, 5G gNodeB)***	Set this parameter to **OFF(OFF)**.	Network plan (negotiation not required)
Dormancy Switch	***DORMANCYSW (LTE eNodeB, 5G gNodeB)***	Set this parameter as required for RF modules.	Network plan (negotiation not required)
a: For the pRRU565*xAB*/pRRU575*xAB* (*x* is a digit, and *A* and *B* are letters), set **RRU type** to **EPRRU**. For other pRRU models, set **RRU type** to **MPMU**.			

CPRI MUX in Separate-MPT

Obey the following rules when multimode RF modules work in multimode scenarios or the BBP of the converged party provides optical fiber connections to a single-mode RF module:

- RF units are configured and monitored on a per mode basis.
- RF units working in the same mode must have the same configuration. Otherwise, ALM-26274 Inter-System Board Object Configuration Conflict will be reported.
- RF units must use the same parameter settings as listed in **Table 4-13**.

Table 4-13 Configuration of common parameters

Parameter Name in UMTS/LTE/NR	Parameter ID in UMTS/LTE/NR
Topo Type	***TT (LTE eNodeB, 5G gNodeB)***
Cabinet No.	***CN (LTE eNodeB, 5G gNodeB)***
Subrack No.	***SRN (LTE eNodeB, 5G gNodeB)***
Slot No.	***SN (LTE eNodeB, 5G gNodeB)***
RRU type	***RT (LTE eNodeB, 5G gNodeB)***
RF Unit Working Mode	***RS (LTE eNodeB, 5G gNodeB)***

The RHUB and pRRU (or RRU) configuration of the converged party takes effect only after the RHUB and pRRU (or RRU) data of the converging party has been configured.

Before activating CPRI MUX, configure the **RRUCHAIN** MO on the converged party (UMTS or LTE). The key parameters are listed in **Table 4-14**.

Table 4-14 Key parameters for configuring RRU chains

Parameter Name	Parameter ID	Setting Notes	Data Source
Access Type	***AT (LTE eNodeB, 5G gNodeB)***	Set this parameter to **PEERPORT**.	Engineering design
Head Cabinet No.	***HCN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the cabinet number of the baseband processing unit or interface board that provides the peer CPRI port.	Engineering design
Head Subrack No.	***HSRN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the subrack number of the baseband processing unit or interface board that provides the peer CPRI port.	Engineering design
Head Slot No.	***HSN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the slot number of the baseband processing unit or interface board that provides the peer CPRI port.	Engineering design

Parameter Name	Parameter ID	Setting Notes	Data Source
Head Port No.	***HPN (LTE eNodeB, 5G gNodeB)***	Set this parameter to the optical port number of the peer CPRI port.	Engineering design
Local Slot No.	***LSN (LTE eNodeB, 5G gNodeB)***	If a BBU3900 is used, the slot number does not need to be configured for the converged board. In this case, set this parameter to **255**. If a BBU3910 is used and uses UL separate-MPT CPRI MUX, set this parameter to the slot number of the converged board.	Equipment planning
CPRI Line Rate	***CR (LTE eNodeB, 5G gNodeB)***	Set this parameter to **AUTO(Auto)**.	Engineering design
User-Defined Rate Negotiation Switch	***USERDEFRATENEGOSW (LTE eNodeB, 5G gNodeB)***	This parameter specifies the user-defined rate negotiation switch. If this switch is turned on, users can specify the **Bit Rate Set** and **Rate Change Period** parameters.	Engineering design

CPRI MUX in Co-MPT

In co-MPT scenarios, if the networking meets the requirements in **Constraints** (BBU3900)/ **Constraints** (BBU3910)/ **Constraints** (BBU5900/BBU5910), CPRI MUX is supported by default. Activating this feature does not require software configurations.

4.4.2.2 Using MML Commands

Activation Command Examples

UL separate-MPT on the BBU3910 is used as example.

```
//Adding an RHUB chain on the converging party
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=0, HSN=2, HPN=0;
//Adding an RHUB to the RHUB chain on the converging party
ADD RHUB: CN=0, SRN=80, RCN=0, PS=0, RN="RHUB1";
//Adding a pRRU/RRU chain on the converging party
ADD RRUCHAIN: RCN=81, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=80, HSN=0, HPN=0, CR=AUTO,
USERDEFRATENEGOSW=OFF;
//Adding a pRRU/RRU to the pRRU/RRU chain on the converging party
ADD RRU: CN=0, SRN=81, SN=0, TP=BRANCH, RCN=81, PS=0, RT=MPMU, RS=UL, RN="prru1", RXNUM=2,
TXNUM=2, MNTMODE=NORMAL, RFTXSIGNDETECTSW=OFF, DORMANCYSW=OFF;
//Adding an RHUB chain on the converged party (UMTS or LTE)
ADD RRUCHAIN: RCN=1, TT=CHAIN, BM=COLD, AT=PEERPORT, HSRN=0, HSN=2, HPN=0, LSN=3,
CR=AUTO, USERDEFRATENEGOSW=OFF;
//Adding an RHUB to the RHUB chain on the converged party
ADD RHUB: CN=0, SRN=80, RCN=1, PS=0, RN="RHUB1";
```

```
//Adding a pRRU/RRU chain on the converged party (UMTS or LTE)
ADD RRUCHAIN: RCN=81, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=80, HSN=0, HPN=0, CR=AUTO,
USERDEFRATENEGOSW=OFF;
//Adding a pRRU/RRU to the pRRU/RRU chain on the converged party (UMTS or LTE)
ADD RRU: CN=0, SRN=81, SN=0, TP=BRANCH, RCN=81, PS=0, RT=MPMU, RS=UL, RN="prru1", RXNUM=2,
TXNUM=2, MNTMODE=NORMAL, RFTXSIGNDETECTSW=OFF, DORMANCYSW=OFF;
```

NOTE

Data must be configured the same for RF modules on the converging party and converged party. For detailed parameter settings, see **Table 4-13**. When configuring the parameter **AT** in separate-MPT CPRI MUX, set the local port for the converging party and the peer port for the converged party. For details, see **Table 4-14**.

In separate-MPT CPRI MUX where the CPRI line rate is 9.8 Gbit/s, if the bandwidth compression mode is changed for a cell on a CPRI link, all cells on the CPRI link must be deactivated, reconfigured, and then activated.

Optimization Command Examples

None

Deactivation Command Examples

This feature does not need to be deactivated.

4.4.2.3 Using the MAE-Deployment

Using the MAE-Deployment to Perform Single Configuration

This function can be activated for a single base station on the MAE-Deployment. The following table lists the MOs and parameters to be configured.

For detailed operations, see Feature Configuration Using the MAE-Deployment.

Table 4-15 Parameters for configuring an RHUB chain on the converging party

MO	Parameter Name	Parameter ID
RRUCHAIN	Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	Topo Type	***TT (LTE eNodeB, 5G gNodeB)***
	Access Type	***AT (LTE eNodeB, 5G gNodeB)***
	Head Cabinet No.	***HCN (LTE eNodeB, 5G gNodeB)***
	Head Subrack No.	***HSRN (LTE eNodeB, 5G gNodeB)***
	Head Slot No.	***HSN (LTE eNodeB, 5G gNodeB)***

MO	Parameter Name	Parameter ID
	Head Port No.	***HPN (LTE eNodeB, 5G gNodeB)***

Table 4-16 Parameters for configuring an RHUB on the RHUB chain on the converging party

MO	Parameter Name	Parameter ID
RHUB	Cabinet No.	***CN (LTE eNodeB, 5G gNodeB)***
	Subrack No.	***SRN (LTE eNodeB, 5G gNodeB)***
	Slot No.	***SN (LTE eNodeB, 5G gNodeB)***
	RRU Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	RHUB Position	***PS (LTE eNodeB, 5G gNodeB)***
	RHUB Name	***RN (LTE eNodeB, 5G gNodeB)***

Table 4-17 Parameters for configuring a pRRU chain or an RRU chain on the converging party

MO	Parameter Name	Parameter ID
RRUCHAIN	Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	Topo Type	***TT (LTE eNodeB, 5G gNodeB)***
	Access Type	***AT (LTE eNodeB, 5G gNodeB)***
	Head Cabinet No.	***HCN (LTE eNodeB, 5G gNodeB)***
	Head Subrack No.	***HSRN (LTE eNodeB, 5G gNodeB)***
	Head Slot No.	***HSN (LTE eNodeB, 5G gNodeB)***
	Head Port No.	***HPN (LTE eNodeB, 5G gNodeB)***

Table 4-18 Parameters for configuring a pRRU or an RRU on the pRRU/RRU chain on the converging party

MO	Parameter Name	Parameter ID
RRU	Cabinet No.	***CN (LTE eNodeB, 5G gNodeB)***
	Subrack No.	***SRN (LTE eNodeB, 5G gNodeB)***
	Slot No.	***SN (LTE eNodeB, 5G gNodeB)***
	RRU Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	RRU type	***RT (LTE eNodeB, 5G gNodeB)***
	RF Unit Working Mode	***RS (LTE eNodeB, 5G gNodeB)***

Table 4-19 Parameters for configuring an RHUB chain on the converged party (UMTS or LTE as the converging party)

MO	Parameter Name	Parameter ID
RRUCHAIN	Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	Topo Type	***TT (LTE eNodeB, 5G gNodeB)***
	Access Type	***AT (LTE eNodeB, 5G gNodeB)***
	Head Cabinet No.	***HCN (LTE eNodeB, 5G gNodeB)***
	Head Subrack No.	***HSRN (LTE eNodeB, 5G gNodeB)***
	Head Slot No.	***HSN (LTE eNodeB, 5G gNodeB)***
	Head Port No.	***HPN (LTE eNodeB, 5G gNodeB)***
	Local Slot No.	***LSN (LTE eNodeB, 5G gNodeB)***

Table 4-20 Parameters for configuring an RHUB on the RHUB chain on the converged party

MO	Parameter Name	Parameter ID
RHUB	Cabinet No.	***CN (LTE eNodeB, 5G gNodeB)***
	Subrack No.	***SRN (LTE eNodeB, 5G gNodeB)***
	Slot No.	***SN (LTE eNodeB, 5G gNodeB)***
	RRU Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	RHUB Position	***PS (LTE eNodeB, 5G gNodeB)***
	RHUB Name	***RN (LTE eNodeB, 5G gNodeB)***

Table 4-21 Parameters for configuring a pRRU chain or an RRU chain on the converged party

MO	Parameter Name	Parameter ID
RRUCHAIN	Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	Topo Type	***TT (LTE eNodeB, 5G gNodeB)***
	Access Type	***AT (LTE eNodeB, 5G gNodeB)***
	Head Cabinet No.	***HCN (LTE eNodeB, 5G gNodeB)***
	Head Subrack No.	***HSRN (LTE eNodeB, 5G gNodeB)***
	Head Slot No.	***HSN (LTE eNodeB, 5G gNodeB)***
	Head Port No.	***HPN (LTE eNodeB, 5G gNodeB)***
	Local Slot No.	***LSN (LTE eNodeB, 5G gNodeB)***

Table 4-22 Parameters for configuring a pRRU or an RRU on the pRRU/RRU chain on the converged party

MO	Parameter Name	Parameter ID
RRU	Cabinet No.	***CN (LTE eNodeB, 5G gNodeB)***
	Subrack No.	***SRN (LTE eNodeB, 5G gNodeB)***
	Slot No.	***SN (LTE eNodeB, 5G gNodeB)***
	RRU Chain No.	***RCN (LTE eNodeB, 5G gNodeB)***
	RRU type	***RT (LTE eNodeB, 5G gNodeB)***
	RF Unit Working Mode	***RS (LTE eNodeB, 5G gNodeB)***

Using the MAE-Deployment to Perform Batch Configuration

This feature does not support batch modification on the MAE-Deployment.

4.4.3 Activation Verification

After the configured data is activated, check whether pRRU-/RRU-related alarms are reported on the **Alarm/Event** tab page of the NodeB and eNodeB LMT or MAE. For details about pRRU-/RRU-related alarms, see **4.4.6.1 Related Alarms**.

1. Check whether the status of pRRU/RRU is normal.
   - **Separate-MPT:**

     On the NodeB LMT or MAE of a 3900 series/5900 series base station, run the **DSP BRD** command to check whether pRRUs/RRUs are functioning properly.

     On the eNodeB LMT or MAE of a 3900 series/5900 series base station, run the **DSP BRD** command to check whether pRRUs/RRUs are functioning properly.
   - **Co-MPT:**

     On the LMT or MAE of a 3900 series/5900 series base station, run the **DSP BRD** command to check whether pRRUs/RRUs are functioning properly.
2. Check whether the CPRI line rate of the converging port is the same as planned.
   - **Separate-MPT:** On the LMT or MAE of the converging party of a 3900 series/5900 series base station, run the **DSP CPRIPORT** command to check whether the CPRI line rate of the converging port is the same as planned.

– **Co-MPT:** Run the **DSP CPRIPORT** command to check whether the CPRI line rate of the converging port is the same as planned.

4.4.4 Network Monitoring

N/A

4.4.5 Reconfiguration

For the following reconfiguration in the branch chain (single-link) and branch load sharing networking scenarios, see *RF Unit and Topology Management*.

- Adding an RHUB to a chain
- Adding a pRRU or an RRU to an RHUB
- Removing an RHUB from a chain
- Removing a pRRU or an RRU from an RHUB

4.4.6 Possible Issues

4.4.6.1 Related Alarms

Table 4-23 lists alarms related to CPRI MUX. For details about troubleshooting operations, see *3900 & 5900 Series Base Station Alarm Reference*.

Table 4-23 CPRI MUX-related alarms

Alarm ID	Alarm Name
ALM-26215	Inter-Board Service Link Failure
ALM-26235	RF Unit Maintenance Link Failure
ALM-26270	Inter-System Communication Failure
ALM-26272	Inter-System RF Unit Parameter Settings Conflict
ALM-26274	Inter-System Board Object Configuration Conflict
ALM-26277	Inter-System Control Rights Conflict
ALM-26278	RF Unit Working Mode and Board Capability Mismatch
ALM-28203	Local Cell Unusable
ALM-28206	Local Cell Capability Decline
ALM-29240	Cell Unavailable
ALM-29243	Cell Capability Degraded

NOTE

In separate-MPT CPRI MUX, if a UMTS cell fails to be activated and EVT-28200 Iub Common Procedure Failure Indication is reported, check whether the UL cell configurations exceed the specifications of a single Ethernet cable. If they do, reconfigure the UL cell specifications. If they do not, run the **BLK CPRIPORT** command to block the CPRI_E/CPRI_O port on the branch chain of the RHUB and then the **UBL CPRIPORT** command to unblock the port.

4.4.6.2 MML Commands for Maintenance

Converging and converged parties have separate O&M platforms in separate-MPT scenarios, and share one O&M platform in co-MPT scenarios.

NOTE

After CPRI MUX is implemented, alarms will be reported if any exceptions occur. Clear the alarms according to the alarm reference.

Table 4-24 MML commands for maintenance

MML Command	Application Scenario	Support by the Converging Party	Support by the Converged Party (Separate-MPT only)
DSP RRUCHAIN	Use this command to check the number of pRRUs/RRUs and RHUBs carried on a chain or ring over an optical port on the converging party and number of RHUB cascading levels.	Fully supported	Fully supported
DSP RRUCHAINPHYTOPO	Use this command to check the connections and physical information of the pRRUs/RRUs and RHUBs connected to the optical port on the panel of the converging party.	Fully supported	Fully supported

MML Command	Application Scenario	Support by the Converging Party	Support by the Converged Party (Separate-MPT only)
DSP CPRIPORT	Use this command to check the real-time dynamic rates and other information of the optical port on the panel of the converging party. Use this command to check whether the optical port is in convergence state.	Fully supported	Partially supported: Use this command to check data rates on the CPRI port and the port convergence status.
DSP CPRILBR[a]	Use this command to check the current line rate of the specified chain or ring, maximum line rate of the baseband processing unit, and maximum line rate of each RHUB.	Fully supported	Partially supported: Use this command to check the line rate on the head and the end of the chain or ring. Other information is displayed as NULL.

MML Command	Application Scenario	Support by the Converging Party	Support by the Converged Party (Separate-MPT only)
STR CPRILBRNEG[b]	Use this command to start negotiation of the line rate on the optical port on the panel. Run this command if the RHUB chain cannot be set up or the chain cannot operate at the maximum line rate due to changes in parameter settings of the RHUB chain. Executing this command causes link disconnection, interrupting services.	Supported	Not supported
MOD RRUCHAIN	Use this command to set breakpoints between an RHUB and pRRUs/RRUs.	Supported	Not supported
L: DSP CELL U: DSP ULOCELL R: DSP RFACELL	Use this command to check the cell status. Cell setup fails and cell capability degrades when CPRI bandwidth is insufficient.	Supported	Supported
a and b: For the pRRU565*xAB*/pRRU575*xAB* (*x* is a digit, and *A* and *B* are letters), after configuration for the pRRU is complete and the pRRU is running properly, CPRI line rate negotiation will be automatically performed for the branch chain. Therefore, you do not need to run the **STR CPRILBRNEG** command. You can run the **DSP CPRIPORT** command to query the CPRI rates supported by the branch chains where the pRRUs reside.			

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see the *Glossary*.

8 Reference Documents

1. 3900 & 5900 Series Base Station Technical Description
2. 3900 & 5900 Series Base Station Installation Guide
3. 3900 & 5900 Series Base Station Maintenance Guide
4. RF Unit and Topology Management Feature Parameter Description
5. 3900 & 5900 Series Base Station MML Command Reference
6. 3900 & 5900 Series Base Station Alarm Reference
7. CPRI MUX Feature Parameter Description
8. LampSite BBU Technical Specifications